

# SAP BTP Documentation - GENERAL

## BTP Documentation: general

URL: <https://help.sap.com/docs/btp/best-practices/basic-platform-concepts>

Title: Basic Platform Concepts | SAP Help Portal

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Basic Platform Concepts

SAP BTP offers users the ability to turn data into business value, compose end-to-end business processes, and build and extend SAP applications quickly.

SAP Business Technology Platform is built on a multi-cloud foundation, which lets you choose from different infrastructures and runtimes. The services and solutions of SAP BTP are available on multiple cloud infrastructure providers, and it supports different runtimes, such as Cloud Foundry, ABAP, and Kyma, as well as multiple regions, and a broad choice of programming languages.

The central point of entry to the platform is the SAP BTP cockpit, where you can access your accounts and applications and manage all activities associated with them.

Check out the SAP Discovery Center to get an overview on the available SAP BTP services, discover missions that guide you step-by-step, and explore reference architectures.

Overview of Global Accounts, Directories, and Subaccountsà^

The following diagram depicts the relationship between a global account, its directories, subaccounts, runtime environments, regions, entitlements, and quotas.

Administrative Tasksà^

Level

Administrative Tasks

Global Account

Global accounts are region- and runtime-independent.

Usually, for each commercial model (licence type), you get a separate global account. It is however possible to combine two commercial models in one global account. See Commercial Models.

Appoint at least one person as administrator. The administrator is responsible for adding new subaccounts, adding members to a global account, and managing the entitlements.

You purchase entitlements for each global account (according to your commercial model). The administrator of the global account distributes quotas to the individual subaccounts.

Directory (optional)

Directories are groups of subaccounts that you can manage, operate, and analyze together.

Appoint at least one person as administrator. The administrator is responsible for adding new subaccounts, managing members, and managing entitlements.

Subaccount

Each subaccount runs in exactly one region. Within a subaccount, you can enable multiple, different runtime environments.

Appoint at least one person as administrator. The administrator is responsible for adding new members to the subaccount and assigning their business roles.

Recommendation

For each administrator, we recommend that you also appoint at least one substitute administrator. If the main administrator leaves the company or is unavailable, it's important that you have someone who's available to take over these tasks.

For more information, see Account Model and Setting Up Your Account Model.

Capabilities and Servicesà^

SAP BTP offers fast in-memory processing, sustainable, agile solutions and services to integrate data and extend applications, and fully embedded analytics and intelligent technologies.

Services enable, facilitate, or accelerate the development of business applications and other platform services on SAP BTP.

For a complete list of services and capabilities, see Services.

For more information, see [Solutions and Services](#).

#### Entitlements and Quotas<sup>^</sup>

When you purchase an enterprise account, you're entitled to use a specific set of resources, such as the amount of memory that can be allocated to your applications.

On SAP BTP, all external dependencies such as databases, messaging systems, files systems, and so on, are services. In this context, multitenant applications and environments are considered services.

Each service has one or more service plans available. A service plan is the representation of the costs and benefits for a given variant of a particular service. For instance, a database may be configured with various "T-shirt sizes", each of which is a different service plan.

An entitlement is your right to provision and consume a resource. In other words, entitlements are the service plans that you're entitled to use.

A quota represents the numeric quantity of a service plan that you're entitled to consume in your global account and its subaccounts.

Entitlements are either managed at global account or directory level and are distributed from there to subaccounts.

For more information, see [Entitlements and Quotas](#) and [Managing Entitlements and Quotas Using the Cockpit](#).  
[User and Member Management](#)<sup>^</sup>

On SAP BTP, user management takes place at all levels from global account to environment. There are different types of users, such as depending on their roles in the company.

User accounts enable users to log on to SAP BTP, access subaccounts, and to use applications according to the permissions granted to them. We distinguish between two types of users:

Platform users are usually developers, administrators or operators who deploy, administer, and troubleshoot accounts, applications and services on SAP BTP.

Business users use the applications that are deployed to SAP BTP. For example, the end users of SaaS apps or services, such as SAP Build Work Zone, or end users of your custom applications are business users.

Member management refers to managing permissions for platform users. A member is a user who is assigned to an SAP BTP global account or subaccount. Administrators can add users to global accounts and subaccounts and assign roles to them as needed. You can use predefined roles, for example the administrator role for managing subaccount members.

User management refers to managing authentication and authorization for your business users.

For more information, see [User and Member Management](#).

#### Account Model<sup>^</sup>

The SAP BTP cockpit is structured according to global accounts and subaccounts. You can also use directories and labels to organize your account model.

#### Global Accounts

A global account is the realization of a contract you or your company has made with SAP.

A global account is used to manage subaccounts, members, entitlements and quotas. You receive entitlements and quotas to use platform resources per global account and then distribute the entitlements and quotas to the subaccount for actual consumption. There are two types of commercial models for global accounts: consumption-based model and subscription-based model. See [Commercial Models](#) .

Global accounts are region- and environment-independent. Within a global account, you manage all of your subaccounts, which in turn are specific to one region.

#### Subaccounts

In subaccounts, you deploy applications, use services, and manage your subscriptions. You can use them to manage members, authorizations, and entitlements.

A global account can contain one or more subaccounts in which you deploy applications, use services, and manage your subscriptions. Subaccounts in a global account are independent from each other. This is important to consider with respect to security, member management, data management, data migration, integration, and so on, when you plan your landscape and overall architecture.

Each subaccount is associated with a region, which is the physical location where applications, data, or services are hosted. The specific region is relevant when you deploy applications and access the SAP BTP cockpit using the corresponding cockpit URL. The region assigned to your subaccount doesn't have to be directly related to your location. You could be located in the United States, for example, but operate your subaccount in Europe.

The entitlements and quotas that have been purchased for a global account have to be assigned to the individual subaccounts.

Global accounts and subaccounts are completely independent of user accounts. For more information, see [User and Member Management](#).

#### Subaccounts, Cloud Foundry Orgs and Kyma Clusters

When you enable the Cloud Foundry environment in one of your subaccounts, the system automatically creates a Cloud Foundry org for you. The subaccount and the org have a 1:1 relationship and the same navigation level in the cockpit (even though they may have different names). You can create spaces within that Cloud Foundry org. Spaces let you further break down your account model and use services and functions in the Cloud Foundry environment.

For more information about Cloud Foundry orgs and spaces, see the Cloud Foundry documentation at <https://docs.cloudfoundry.org/concepts/roles.html>.

When you enable the Kyma environment in one of your subaccounts, you can provision one or more Kubernetes clusters equipped with the latest version of the open-source project "Kyma". The subaccount and the Kyma cluster have a 1:n relationship - you can choose to create more than one cluster right away, or add clusters later.

Within each Kyma cluster, you can create namespaces to organize your workloads and control access to resources. Namespaces let you further break down your account model and use services and functions in the Kyma environment. For more information, see the Kubernetes documentation at <https://kubernetes.io/docs/concepts/overview/working-with-objects/namespaces/>.

#### Directories

With directories, you can organize and manage your subaccounts according to your technical and business needs.

A directory can contain directories and subaccounts to create a hierarchy. Using directories to group other directories and subaccounts is optional - you can still create subaccounts directly under your global account. You can create a hierarchical structure that is up to 7 levels deep. The highest level of a given path is always the global account and the lowest is a subaccount, which means that you can have up to 5 levels of directories between the global account and the lowest level subaccount.

Directories enable you to:

- Group and filter directories and subaccounts

- Monitor usage and costs for contracts that use the consumption-based commercial model

Optionally, you can also enable the following features in your directories:

**Manage Entitlements:** Enables the assignment of a quota for services and applications to the directory from the global account quota, for distribution to the directory's subaccounts.

When you assign entitlements to a directory, you express the entitlements and maximum quota that can be distributed across its children subaccounts. You also have the option to choose the auto-assignment of a set amount of quota to all subaccounts created or moved to that directory. Subaccounts that are already in the directory when you select that option will not be auto-assigned quota.

#### Note

If you've enabled the Manage Entitlements feature for a given directory, you must first assign the necessary entitlements and maximum allowed quota from the global account to that directory. Then you can distribute this "reserved" quota to any of the directory's child subaccounts.

**Manage Authorizations:** Enables authorization management for the directory. For example, it allows certain users to manage directory entitlements. You can only use this feature in combination with the Manage Entitlements feature.

#### Labels

Labels are user-defined words or phrases that you can assign to various entities in SAP BTP to categorize them in your global account, to identify them more easily. They let you organize and filter your directories, subaccounts, instances and subscriptions within your global account.

For more information, see [Labels](#).

#### Related Information

- [Account Model](#)

- [Managing Global Accounts Using the Cockpit](#)

- [Managing Directories Using the Cockpit](#)

## Managing Subaccounts Using the Cockpit

### Setting Up Your Account Model

#### Commercial Models

SAP BTP offers two different commercial models for enterprise accounts.

**Consumption-based commercial model:** Your organization receives access to all current and future services that are eligible for this model. You have complete flexibility to turn services on and off and to switch between services as your business requires throughout the duration of your contract. This commercial model is available in the following flavors: SAP BTP Enterprise Agreement (SAP BTPEA), Cloud Platform Enterprise Agreement (CPEA), and Pay-As-You-Go for SAP BTP.

For more information, see [What Is the Consumption-Based Commercial Model?](#)

**Subscription-based commercial model:** Your organization subscribes only to the services that you plan to use. You can then use these services at a fixed cost, irrespective of consumption.

For more information, see [What Is the Subscription-Based Commercial Model?](#)

#### Note

You can use both commercial models, either in separate global accounts or in the same global account depending on your business needs. Contact your SAP account executive or sales representative for more information. Note however that you cannot mix more than one flavor of the consumption-based commercial model in the same global account.

For information about service availability, prices, and estimators, see <https://www.sap.com/products/technology-platform/solutions.html> and <https://www.sap.com/products/technology-platform/integration-suite/pricing.html>.

You can also view the service catalog via the SAP Discovery Center.

#### Environments

Environments constitute the actual platform-as-a-service offering of SAP BTP that allows for the development and administration of business applications. Environments are anchored in SAP BTP on subaccount level. Each environment comes equipped with specific tools, technologies, and runtimes that you need to build applications. So a multi-environment subaccount is your single address to host a variety of applications and offer diverse development options. One advantage of using different environments in one subaccount is that you only need to manage users, authorizations, and entitlements once per subaccount, and thus grant more flexibility to your developers.

You can use different environments within the same subaccount: Choose multi-environment subaccounts to use Cloud Foundry, ABAP, and Kyma within the same subaccount. However, inside such a multi-environment subaccount, you cannot use Neo. To use Neo, you create a separate Neo subaccount. Neo cannot be combined with other environments in the same subaccount.

SAP offers the following environments:

#### Cloud Foundry Environment

The Cloud Foundry environment enables you to develop new business applications and business services, supporting multiple runtimes, programming languages, libraries, and services. You can leverage a multitude of buildpacks, including community innovations and self-developed buildpacks.

#### ABAP Environment

The ABAP environment allows you to create extensions for ABAP-based products, such as SAP S/4HANA Cloud, and develop new cloud applications. You can transform existing ABAP-based custom code or extensions to the cloud.

To plan and set up your landscape and lifecycle management in the ABAP environment, see [ABAP Lifecycle Management](#).

#### Kyma Environment

SAP BTP, Kyma runtime provides a fully managed cloud-native Kubernetes application runtime based on the open-source project "Kyma". Based on modular building blocks, Kyma runtime includes all the necessary capabilities to simplify the development and to run enterprise-grade cloud-native applications.

#### Neo Environment

The Neo environment lets you develop HTML5, Java, and SAP HANA extended application services (SAP HANA XS) applications. You can also use the UI Development Toolkit for HTML5 (SAPUI5) to develop rich user interfaces for modern web-based business applications.

If you consider creating Neo subaccounts, see [Migrating from the Neo Environment to the Multi-Cloud Foundation for SAP BTP \(Cloud Foundry and Kyma\)](#) to learn about the Neo environment and the multi-cloud

foundation.

#### Environment Instances

To actually use an environment in a subaccount, you must enable it, which creates an instance of that environment. There are several ways to create environment instances:

In the SAP BTP cockpit, on the subaccount overview, choose Enable.

In the SAP BTP cockpit, under Service Marketplace. Here, you get more information, such as the available plans and links to further information.

Using the `btp` CLI command `btp create accounts/environment-instance`

#### Related Information

SAP BTP, Serverless Runtime

Consuming SAP BTP Services from Various Environments

#### Regions

You can deploy applications in different regions. Each region represents a geographical location (for example, Europe, US East) where applications, data, or services are hosted.

Regions are provided either by SAP or by our Infrastructure-as-a-Service (IaaS) partners Amazon Web Services (AWS), Microsoft Azure, Google Cloud, and Alibaba Cloud. The third-party region providers operate the infrastructure layer of the regions, whereas SAP operates the platform layer and Cloud Foundry.

A region is chosen at the subaccount level. For each subaccount, you select exactly one region (that is one data center).

For more information, see Regions.

#### Selecting a Region

When deciding on the location of your Platform as a Service (PaaS), consider existing Software as a Service (SaaS) and Infrastructure as a Service (IaaS) and try to locate it close to those or even in the same data center.

You can also optimize application performance (response time, latency) by selecting a region that's geographically close to your users. However, the selection of a region also depends on many other factors:

Check the availability of specific services in the individual regions.

Ensure that you comply with security requirements, such as country- or industry-specific data privacy regulations.

Consider the location of other cloud offerings you're using. You might have to locate your solutions in the same data center.

For a complete overview of the availability of services in the different regions, see Services.

#### Note

To work with the Kyma environment, you need to specify the region for both your subaccount and the cluster.

Note that there is a number of regions available within each plan. For details, see Regions for the Kyma Environment.

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URL: <https://help.sap.com/docs/btp/best-practices/creating-governance-model>

Title: *Creating a Governance Model* | SAP Help Portal

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#### Creating a Governance Model

One of the first and most important steps of your journey to the cloud is to establish an appropriate organizational setup and corresponding governance model. A clear and well-thought-out organizational setup makes it easier for your employees to adopt agile processes.

We also recommend that before you begin your development project, you create an onboarding process and a knowledge transfer process.

For example:

Define team setups (including IT support roles and responsibilities)

Create an onboarding process for development projects

Create a knowledge transfer process for the involved teams

Define support processes, operations documentation, and involved tools

Define activities to ramp up resources and implement changes (support processes, tools, documentation)

URL: <https://help.sap.com/docs/btp/btp-admin-guide/350356d1dc314d3199dca15bd2ab9b0e.html>

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Regions

You can deploy applications in different regions. Each region represents a geographical location (for example, Europe, US East) where applications, data, or services are hosted.

About SAP BTP Regionsà^

Regions are provided either by SAP or by our Infrastructure-as-a-Service (IaaS) partners Amazon Web Services (AWS), Microsoft Azure, Google Cloud, and Alibaba Cloud. The third-party region providers operate the infrastructure layer of the regions, whereas SAP operates the platform layer and Cloud Foundry.

For an overview of all available regions, see SAP Discovery Center:

Selecting a Regionà^

A region is chosen at the subaccount level. For each subaccount, you select exactly one region. The selection of a region is dependent on many factors: For example, application performance (response time, latency) can be optimized by selecting a region close to the user. For more information, see Selecting a Region.

Deploying Applications in Regionsà^

When deploying applications, consider that a subaccount is associated with a particular region and that this is independent of your own location. You may be located in the United States, for example, but operate your subaccount in a region in Europe. For more information on subaccounts, see Subaccounts.

To deploy an application in more than one region, execute the deployment separately for each host. For more information, see Deploy an Application.

Within a region, there can be multiple instances of the SAP BTP, Cloud Foundry environment. When creating a subaccount, SAP BTP automatically assigns the account to a specific instance of the environment. Several subaccounts of the same global account can be located in different datacenters in one region. This also affects the format of the API endpoint URL that is displayed in the cockpit after enabling Cloud Foundry in your subaccount. There are two possible formats for the API endpoint URL, either displayed with or without an index. Here's an example for eu10:

Example

<https://api.cf.eu10.hana.ondemand.com>

<https://api.cf.eu10-<XXX>.hana.ondemand.com>

In both cases, the subaccount is located in the region eu10. The differences in the URLs are only an indicator of technical details on the side of SAP BTP and do not affect the functionality of your applications. For information on enabling Cloud Foundry, see Create Orgs.

High Availabilityà^

SAP has a number of processes in place to support resilience in SAP BTP, and provides different offerings so that you can support the high availability of your applications. For more information, see Resilience, High Availability, and Disaster Recovery.

EU Accessà^

Some customer contracts include EU Access, which restricts processing of personal data to EEA/Switzerland. If the global account is marked with EU Access, the actual EU Access compliance status of subaccounts will be displayed during creation of subaccounts.

Note

If you require a subaccount with EU Access, make sure to select a provider and region where EU Access is available. Regions that allow for EU Access are labeled as such in the map view filter in the SAP Discovery Center.

For some services, EU Access is generally not available, not even if the provider and region support EU Access.

Related Informationà^

Regions and API Endpoints Available for the Cloud Foundry Environment

Regions and API Endpoints for the ABAP Environment

URL: <https://help.sap.com/docs/btp/btp-admin-guide/7047eb4a15a84ac7be3c8612179e6d1f.html>

Title: What Is the Consumption-Based Commercial Model? | SAP Help Portal

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What Is the Consumption-Based Commercial Model?

Note

The use of the consumption-based commercial model is subject to its availability in your country or region. With the consumption-based model, your organization purchases an entitlement to all current and future SAP BTP services that are eligible for this model. Throughout the duration of your contract, you have complete flexibility to turn services on and off and to switch between services as your business requires.

The consumption-based commercial model is available in the following flavors: the SAP BTPEA (SAP BTP Enterprise Agreement), the CPEA (Cloud Platform Enterprise Agreement), and Pay-As-You-Go for SAP BTP. Each option is suited to different business situations and levels of financial commitment, as described in the table below. For additional information and clarifications, please contact your account executive.

SAP BTPEA

Your organization makes a prepaid investment in cloud credits for the contract duration with an annual commitment to consume SAP BTP services.

This model is suitable for customers that have well-established and planned use cases, and who want the flexibility of turning services on and off, and switching between services, without the commitment of being tied to a single service throughout the duration of the contract.

You receive a monthly balance statement that provides information about the usage consumption of each service and the corresponding costs. The total monthly cost is deducted from your cloud credits balance.

This model has a minimum investment entry, and volume-based discounts are available.

You are billed annually in advance. Any overages are billed in arrears at list price.

You can top up your cloud credits at any time to prevent overages.

You get access to new SAP BTP services that are added to the consumption-based service catalog.

CPEA

Note

Certain restrictions apply. See the Frequently Asked Questions section in <https://www.sap.com/products/technology-platform/pricing.html>.

Your organization makes a prepaid investment in cloud credits for the contract duration with an annual commitment to consume SAP BTP services.

This model is suitable for customers that have well-established and planned use cases, and who want the flexibility of turning services on and off, and switching between services, without the commitment of being tied to a single service throughout the duration of the contract.

You receive a monthly balance statement that provides information about the usage consumption of each service and the corresponding costs. The total monthly cost is deducted from your cloud credits balance.

This model has a minimum investment entry, and volume-based discounts are available.

You are billed annually in advance. Any overages are billed in arrears at list price.

You can top up your cloud credits at any time to prevent overages.

Pay-As-You-Go for SAP BTP

You have the same access to all the services that are available in SAP BTPEA, but with a highly flexible zero-commitment model – you pay nothing upfront and there is no minimum usage requirement or annual commitment.

You pay only for the SAP BTP services that you want, when you use them.

You are billed monthly in arrears.

Service charges are non-discountable.

This low-risk model is suitable for customers with use cases that are not well defined, and are interested in running a proof-of-concept in a productive environment. This model provides the flexibility of turning services on and off, and switching between services, as needed throughout the duration of the contract.

A seamless transition to the CPEA model is available, on the condition that you have no other CPEA-based global accounts.

If you want information about the Pay-As-You-Go for SAP BTP for cloud test, demo, and development commercial model, go to the SAP Partner Portal.

#### Note

In global accounts that use the consumption-based commercial model, a default runtime memory quota of 200 GB for the standard plan is automatically assigned to every subaccount. This quota represents a limit on how much runtime memory the applications in the subaccount can use at any given time. When you use the standard plan for SAP BTP, Cloud Foundry runtime in the consumption-based commercial model, you're billed based on the amount of runtime memory that has actually been consumed. For more information, see [Service Plans and Metering for Cloud Foundry Runtime and Consumption Monitoring](#).

A global account administrator can change the default quota as described in (optional) [Adjusting Runtime Memory Quota for Cloud Foundry Runtime \(Consumption-Based\)](#).

#### Tip

You can monitor costs and service usage throughout the contract period. See [Monitoring Usage and Consumption Costs in Your Global Account](#).

For information about eligible services and pricing, see <https://www.sap.com/products/technology-platform/pricing.html>, or access the SAP BTP service catalog via the SAP Discovery Center. The SAP BTP service catalog allows you to identify service availability per data center and to determine licensing model compatibility per available service plan.

As part of the free tier model for SAP BTP, some services offer free service plans that allow you to try out these services without any additional charge within your SAP BTPEA, CPEA, or Pay-As-You-Go for SAP BTP global account. For more information, see [Using Free Service Plans](#).

To find frequently asked questions about the consumption-based commercial model, see the [Frequently Asked Questions](#) section in <https://www.sap.com/products/technology-platform/pricing.html>.

#### Note

Depending on your business needs, you can combine the consumption-based commercial model with the subscription-based commercial model in the same global account, or you can use both commercial models in the same global account. Contact your SAP account executive or sales representative for more information. Note however that you cannot mix more than one flavor of the consumption-based commercial model in the same global account.

#### Related Information

[What Is the Subscription-Based Commercial Model?](#)

[Monitoring Usage and Consumption Costs in Your Global Account](#)

[View Subaccount Usage Analytics](#)

[Managing Global Accounts Using the Cockpit](#)

[Managing Directories Using the Cockpit](#)

[Managing Subaccounts Using the Cockpit](#)

[Managing Entitlements and Quotas Using the Cockpit](#)

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Cloud Foundry Environment

The Cloud Foundry environment allows you to create polyglot cloud applications in Cloud Foundry. It contains the SAP BTP, Cloud Foundry runtime service, which is based on the open-source application platform managed by the Cloud Foundry Foundation.

The Cloud Foundry environment enables you to develop new business applications and business services, supporting multiple runtimes, programming languages, libraries, and services. You can leverage a multitude of buildpacks, including community innovations and self-developed buildpacks. It also integrates with SAP HANA extended application services, advanced model.

For more information about Cloud Foundry, see the official Cloud Foundry documentation at <https://docs.cloudfoundry.org/>.

Supported and Unsupported Cloud Foundry Features

Find out which Cloud Foundry features the Cloud Foundry environment on SAP BTP supports and doesn't support.

SAP BTP-Specific Configurations

Learn about technical configurations for Cloud Foundry that are specific to SAP BTP.

Availability Zones in the Cloud Foundry Environment

The Cloud Foundry environment follows the recommendations of our partner IaaS providers by leveraging the availability zones (AZ) concept.

Additional Information About Cloud Foundry

Links to additional information about Cloud Foundry that is useful to know but not necessarily directly connected to the SAP BTP, Cloud Foundry environment.

Service Plans and Metering for Cloud Foundry Runtime

This page explains the relationship between the service plans in the SAP Discovery Center and those in the SAP BTP cockpit, and provides information to help you understand how the service is billed.

Related Informationà^

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#### Administrative Tasks

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For more information, see Account Model and Setting Up Your Account Model.

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For more information, see [Solutions and Services](#).

#### Entitlements and Quotas<sup>^</sup>

When you purchase an enterprise account, you're entitled to use a specific set of resources, such as the amount of memory that can be allocated to your applications.

On SAP BTP, all external dependencies such as databases, messaging systems, files systems, and so on, are services. In this context, multitenant applications and environments are considered services.

Each service has one or more service plans available. A service plan is the representation of the costs and benefits for a given variant of a particular service. For instance, a database may be configured with various "T-shirt sizes", each of which is a different service plan.

An entitlement is your right to provision and consume a resource. In other words, entitlements are the service plans that you're entitled to use.

A quota represents the numeric quantity of a service plan that you're entitled to consume in your global account and its subaccounts.

Entitlements are either managed at global account or directory level and are distributed from there to subaccounts.

For more information, see [Entitlements and Quotas](#) and [Managing Entitlements and Quotas Using the Cockpit](#).  
[User and Member Management](#)<sup>^</sup>

On SAP BTP, user management takes place at all levels from global account to environment. There are different types of users, such as depending on their roles in the company.

User accounts enable users to log on to SAP BTP, access subaccounts, and to use applications according to the permissions granted to them. We distinguish between two types of users:

Platform users are usually developers, administrators or operators who deploy, administer, and troubleshoot accounts, applications and services on SAP BTP.

Business users use the applications that are deployed to SAP BTP. For example, the end users of SaaS apps or services, such as SAP Build Work Zone, or end users of your custom applications are business users.

Member management refers to managing permissions for platform users. A member is a user who is assigned to an SAP BTP global account or subaccount. Administrators can add users to global accounts and subaccounts and assign roles to them as needed. You can use predefined roles, for example the administrator role for managing subaccount members.

User management refers to managing authentication and authorization for your business users.

For more information, see [User and Member Management](#).

#### Account Model<sup>^</sup>

The SAP BTP cockpit is structured according to global accounts and subaccounts. You can also use directories and labels to organize your account model.

#### Global Accounts

A global account is the realization of a contract you or your company has made with SAP.

A global account is used to manage subaccounts, members, entitlements and quotas. You receive entitlements and quotas to use platform resources per global account and then distribute the entitlements and quotas to the subaccount for actual consumption. There are two types of commercial models for global accounts: consumption-based model and subscription-based model. See [Commercial Models](#) .

Global accounts are region- and environment-independent. Within a global account, you manage all of your subaccounts, which in turn are specific to one region.

#### Subaccounts

In subaccounts, you deploy applications, use services, and manage your subscriptions. You can use them to manage members, authorizations, and entitlements.

A global account can contain one or more subaccounts in which you deploy applications, use services, and manage your subscriptions. Subaccounts in a global account are independent from each other. This is important to consider with respect to security, member management, data management, data migration, integration, and so on, when you plan your landscape and overall architecture.

Each subaccount is associated with a region, which is the physical location where applications, data, or services are hosted. The specific region is relevant when you deploy applications and access the SAP BTP cockpit using the corresponding cockpit URL. The region assigned to your subaccount doesn't have to be directly related to your location. You could be located in the United States, for example, but operate your subaccount in Europe.

The entitlements and quotas that have been purchased for a global account have to be assigned to the individual subaccounts.

Global accounts and subaccounts are completely independent of user accounts. For more information, see [User and Member Management](#).

#### Subaccounts, Cloud Foundry Orgs and Kyma Clusters

When you enable the Cloud Foundry environment in one of your subaccounts, the system automatically creates a Cloud Foundry org for you. The subaccount and the org have a 1:1 relationship and the same navigation level in the cockpit (even though they may have different names). You can create spaces within that Cloud Foundry org. Spaces let you further break down your account model and use services and functions in the Cloud Foundry environment.

For more information about Cloud Foundry orgs and spaces, see the Cloud Foundry documentation at <https://docs.cloudfoundry.org/concepts/roles.html>.

When you enable the Kyma environment in one of your subaccounts, you can provision one or more Kubernetes clusters equipped with the latest version of the open-source project "Kyma". The subaccount and the Kyma cluster have a 1:n relationship - you can choose to create more than one cluster right away, or add clusters later.

Within each Kyma cluster, you can create namespaces to organize your workloads and control access to resources. Namespaces let you further break down your account model and use services and functions in the Kyma environment. For more information, see the Kubernetes documentation at <https://kubernetes.io/docs/concepts/overview/working-with-objects/namespaces/>.

#### Directories

With directories, you can organize and manage your subaccounts according to your technical and business needs.

A directory can contain directories and subaccounts to create a hierarchy. Using directories to group other directories and subaccounts is optional - you can still create subaccounts directly under your global account. You can create a hierarchical structure that is up to 7 levels deep. The highest level of a given path is always the global account and the lowest is a subaccount, which means that you can have up to 5 levels of directories between the global account and the lowest level subaccount.

Directories enable you to:

- Group and filter directories and subaccounts

- Monitor usage and costs for contracts that use the consumption-based commercial model

Optionally, you can also enable the following features in your directories:

**Manage Entitlements:** Enables the assignment of a quota for services and applications to the directory from the global account quota, for distribution to the directory's subaccounts.

When you assign entitlements to a directory, you express the entitlements and maximum quota that can be distributed across its children subaccounts. You also have the option to choose the auto-assignment of a set amount of quota to all subaccounts created or moved to that directory. Subaccounts that are already in the directory when you select that option will not be auto-assigned quota.

#### Note

If you've enabled the Manage Entitlements feature for a given directory, you must first assign the necessary entitlements and maximum allowed quota from the global account to that directory. Then you can distribute this "reserved" quota to any of the directory's child subaccounts.

**Manage Authorizations:** Enables authorization management for the directory. For example, it allows certain users to manage directory entitlements. You can only use this feature in combination with the Manage Entitlements feature.

#### Labels

Labels are user-defined words or phrases that you can assign to various entities in SAP BTP to categorize them in your global account, to identify them more easily. They let you organize and filter your directories, subaccounts, instances and subscriptions within your global account.

For more information, see [Labels](#).

#### Related Information

- [Account Model](#)

- [Managing Global Accounts Using the Cockpit](#)

- [Managing Directories Using the Cockpit](#)

## Managing Subaccounts Using the Cockpit

### Setting Up Your Account Model

#### Commercial Models

SAP BTP offers two different commercial models for enterprise accounts.

**Consumption-based commercial model:** Your organization receives access to all current and future services that are eligible for this model. You have complete flexibility to turn services on and off and to switch between services as your business requires throughout the duration of your contract. This commercial model is available in the following flavors: SAP BTP Enterprise Agreement (SAP BTPEA), Cloud Platform Enterprise Agreement (CPEA), and Pay-As-You-Go for SAP BTP.

For more information, see [What Is the Consumption-Based Commercial Model?](#)

**Subscription-based commercial model:** Your organization subscribes only to the services that you plan to use. You can then use these services at a fixed cost, irrespective of consumption.

For more information, see [What Is the Subscription-Based Commercial Model?](#)

#### Note

You can use both commercial models, either in separate global accounts or in the same global account depending on your business needs. Contact your SAP account executive or sales representative for more information. Note however that you cannot mix more than one flavor of the consumption-based commercial model in the same global account.

For information about service availability, prices, and estimators, see <https://www.sap.com/products/technology-platform/solutions.html> and <https://www.sap.com/products/technology-platform/integration-suite/pricing.html>.

You can also view the service catalog via the SAP Discovery Center.

#### Environments

Environments constitute the actual platform-as-a-service offering of SAP BTP that allows for the development and administration of business applications. Environments are anchored in SAP BTP on subaccount level. Each environment comes equipped with specific tools, technologies, and runtimes that you need to build applications. So a multi-environment subaccount is your single address to host a variety of applications and offer diverse development options. One advantage of using different environments in one subaccount is that you only need to manage users, authorizations, and entitlements once per subaccount, and thus grant more flexibility to your developers.

You can use different environments within the same subaccount: Choose multi-environment subaccounts to use Cloud Foundry, ABAP, and Kyma within the same subaccount. However, inside such a multi-environment subaccount, you cannot use Neo. To use Neo, you create a separate Neo subaccount. Neo cannot be combined with other environments in the same subaccount.

SAP offers the following environments:

#### Cloud Foundry Environment

The Cloud Foundry environment enables you to develop new business applications and business services, supporting multiple runtimes, programming languages, libraries, and services. You can leverage a multitude of buildpacks, including community innovations and self-developed buildpacks.

#### ABAP Environment

The ABAP environment allows you to create extensions for ABAP-based products, such as SAP S/4HANA Cloud, and develop new cloud applications. You can transform existing ABAP-based custom code or extensions to the cloud.

To plan and set up your landscape and lifecycle management in the ABAP environment, see [ABAP Lifecycle Management](#).

#### Kyma Environment

SAP BTP, Kyma runtime provides a fully managed cloud-native Kubernetes application runtime based on the open-source project "Kyma". Based on modular building blocks, Kyma runtime includes all the necessary capabilities to simplify the development and to run enterprise-grade cloud-native applications.

#### Neo Environment

The Neo environment lets you develop HTML5, Java, and SAP HANA extended application services (SAP HANA XS) applications. You can also use the UI Development Toolkit for HTML5 (SAPUI5) to develop rich user interfaces for modern web-based business applications.

If you consider creating Neo subaccounts, see [Migrating from the Neo Environment to the Multi-Cloud Foundation for SAP BTP \(Cloud Foundry and Kyma\)](#) to learn about the Neo environment and the multi-cloud

foundation.

#### Environment Instances

To actually use an environment in a subaccount, you must enable it, which creates an instance of that environment. There are several ways to create environment instances:

In the SAP BTP cockpit, on the subaccount overview, choose Enable.

In the SAP BTP cockpit, under Service Marketplace. Here, you get more information, such as the available plans and links to further information.

Using the btp CLI command `btp create accounts/environment-instance`

#### Related Information

SAP BTP, Serverless Runtime

Consuming SAP BTP Services from Various Environments

#### Regions

You can deploy applications in different regions. Each region represents a geographical location (for example, Europe, US East) where applications, data, or services are hosted.

Regions are provided either by SAP or by our Infrastructure-as-a-Service (IaaS) partners Amazon Web Services (AWS), Microsoft Azure, Google Cloud, and Alibaba Cloud. The third-party region providers operate the infrastructure layer of the regions, whereas SAP operates the platform layer and Cloud Foundry.

A region is chosen at the subaccount level. For each subaccount, you select exactly one region (that is one data center).

For more information, see Regions.

#### Selecting a Region

When deciding on the location of your Platform as a Service (PaaS), consider existing Software as a Service (SaaS) and Infrastructure as a Service (IaaS) and try to locate it close to those or even in the same data center.

You can also optimize application performance (response time, latency) by selecting a region that's geographically close to your users. However, the selection of a region also depends on many other factors:

Check the availability of specific services in the individual regions.

Ensure that you comply with security requirements, such as country- or industry-specific data privacy regulations.

Consider the location of other cloud offerings you're using. You might have to locate your solutions in the same data center.

For a complete overview of the availability of services in the different regions, see Services.

#### Note

To work with the Kyma environment, you need to specify the region for both your subaccount and the cluster.

Note that there is a number of regions available within each plan. For details, see Regions for the Kyma Environment.

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URL: <https://help.sap.com/docs/btp/btp-admin-guide/data-backups-managed-by-sap>

Title: Data Backups Managed by SAP | SAP Help Portal

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#### Data Backups Managed by SAP

SAP handles the backup and recovery of service data. However, there are some exceptions.

#### Services with Automated Data Backups

Data stored in the following services and components is automatically backed up on a regular basis by SAP:

SAP HANA Cloud: SAP HANA Cloud instances are continually backed up to safeguard your database and ensure that it can be recovered. For more information, see Backup and Recovery.

PostgreSQL on SAP BTP, Hyperscaler Option: SAP keeps backups of your PostgreSQL database instances for a retention period of 14 days. You can restore a database instance to any point in time within the backup retention period by creating a new database instance. For more information, see Restore Database Instance to a Specified Time.

#### Services Without Automated Data Backups

The following services don't currently provide any backup and restore features:

Redis on SAP BTP, hyperscaler option: This service doesn't support persistence.

Object Store on SAP BTP: There's no backup and restore feature provided by the Object Store service.

However, there are additional features that could help with backup and restore in certain scenarios:

Object Versioning allows you to recover from accidental deletion and modification of objects because the hyperscalers keep older versions in case the objects are modified or deleted.

Object Expiration Rules helps with the automatic deletion of older versions.

The prevent deletion feature helps preventing your blob stores from accidental deletion. For more information, see [Prevent Accidental Deletion of AWS S3 Buckets](#) and [Preventing Accidental Deletion of Azure Containers](#)

Runtime

SAP BTP, Cloud Foundry Runtime

For services running in the SAP BTP, Cloud Foundry runtime, the backed-up data is copied to additional availability zones within the same region. Each availability zone represents a physically separate location with its own power supply, network, and cooling. For more information, see [Regions](#).

SAP BTP, Kyma Runtime

SAP BTP, Kyma runtime relies on managed Kubernetes clusters for periodic backups of Kubernetes objects.

Automatic backup doesn't include Kubernetes volumes. For more information, see [Kyma Environment Backup](#).

SAP BTP, Neo Environment

For services running in the SAP BTP, Neo environment, the backed-up data is copied to an availability zone in another region. For more information, see [Regions in the Neo Environment](#).

SAP BTP ABAP Environment

The ABAP environment relies on backups from SAP HANA Cloud. For details, see [Backups](#).

Related Information

SAP BTP Service Configuration Backups Managed by Customers

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URL: <https://help.sap.com/docs/btp/btp-admin-guide/go-live-and-monitor>

Title: [Go Live and Operate | SAP Help Portal](#)

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Go Live and Operate

Once you've deployed, transported, tested, and integrated your application, you can go live.

Going Live

Once you've tested your application successfully and ensured that you are compliant with any applicable security guidelines and compliance regulations, you can go live with your application.

If you've set up a staged development environment, we recommend that your Cloud Administration Team defines specific timeframes during which the Cloud Development Team is allowed to go live with an application.

You can forbid go-live activities during times that a stable landscape is particularly crucial (for example, at the end of each quarter), and allow deployments to your production subaccount only in case of emergencies.

Consider embedding applications with internal end users in the SAP Fiori launchpad using SAP BTP Portal before you go live. This enables your internal target audience to access all applications in one central place and thus improves the app's usability. For more information, see [SAP Cloud Portal Service](#).

Manage Authentication and Authorization

Ensure that the business users of your application are being provided by using SAP ID service or configuring the trust relationship to an external identity provider. They should also get the right authorization by using application-based authorization artifacts provided by the developers. This allows the administrators to create roles, build role collections, and assign these collections to business users and user groups.

For more information, see [Security Administration: Managing Authentication and Authorization](#).

Web Acceleration

If your application's end users are widely spread across different countries or even continents, you can improve your application's load performance by using a Content Delivery Network (CDN). For example, you can use common CDN providers such as Akamai or Cloudflare.

If you include SAP's SAPUI5 library directly from \*.hana.ondemand.com, the SAPUI5 resources are automatically delivered via a CDN. See [Variant for Bootstrapping from Content Delivery Network](#).

Consider the following when using a CDN:

Secure transport: Make sure any access via HTTP will get redirected to a HTTPS connection before loading any content.

Block access based on location: If blocking access from a certain country is required, the client's location data can be used. Keep in mind that this does not guarantee a full blocking, since the client's location data can be changed, or the client could access from another country via a virtual private network (VPN).

Content compression: You can compress your content with gzip before it gets delivered to the client. This improves the performance, especially for clients with a slow connection.

Content caching: In addition to client caching, you can make the CDN provider cache your server's content, so that following requests of the same resources by other clients will be delivered faster. While this can additionally improve the performance, you need to keep in mind the following:

Make sure that you only cache static content. We recommend that you exclude certain files (for example, for UI5 apps, exclude the neo-app.json file) or paths (for example, a route to OData services), to make sure dynamic content is never cached.

You should configure your CDN provider so that it respects the Cache-Control and Expires header of your server.

You should not cache any dynamic header, such as the X-CSRF-Token header, that is used against cross-site request forgery (CSRF).

See SAP Note 2943781 for information about using CDN for on-premise systems.

Operate Centrally with SAP Cloud ALM<sup>^</sup>

Use SAP Cloud ALM to monitor and operate your landscape.

SAP Cloud ALM is included in your SAP Cloud Service subscription containing Enterprise Support, cloud edition, in SAP Enterprise Support and in Product Support for Large Enterprises.

SAP Cloud ALM helps you to implement and operate intelligent cloud and hybrid business solutions. You benefit from an out-of-the-box, native cloud solution running on SAP BTP, designed as the central entry point to manage your complete SAP landscape with content-driven guided implementation and highly automated operations.

With the operations use cases in SAP Cloud ALM, you can handle both SAP BTP-based SaaS applications and your own, customer-built applications.

SAP Cloud ALM for operations provides a personalized and unified operations user experience. You can use purpose-driven monitoring, analysis, and management use cases optimized for certain operations aspects.

SAP Cloud ALM for operations empowers you to understand the health of your SAP business solutions and acts as collaboration platform between lines of business and IT.

Under Supported Solutions, you see which monitoring capabilities are available for which SAP solution with SAP Cloud ALM. Under SAP Solution, filter for BTP and check the supported services from the operations area. Benefit from the powerful Intelligent Event Processing, embedded analytics, and intelligence functionality with a large variety of inbound and outbound integration. Synthetic User Monitoring monitors the performance and availability of web applications from a client-side perspective based on Selenium-based scripts. Business Service Management consolidates all operational events at business service level and shows them in a well-arranged event calendar.

Specifically for Java and Node.js customer-built applications in SAP BTP, Cloud Foundry runtime, there are the Data Collection Instrumentation Libraries based on OpenTelemetry designed to enable the data collection infrastructure in SAP Cloud ALM. With these libraries, you can collect data for the following use cases:

Real User Monitoring: Leverage both front-end and back-end measurements on single request level.

Health Monitoring: Gain insights on application health based on technical metrics.

Integration and Exception Monitoring: Collect and react on exceptions raised by the application.

Job and Automation Monitoring: Monitor SAP Job Scheduling Service jobs.

As SAP Cloud ALM is the central place to monitor your SAP solution landscape, you can integrate it in your existing IT service management processes, for example, you can connect your ticketing system.

Inbound and outbound APIs enable you to integrate data from your own applications and consume the data, for example, for your own analytics purposes. For more information, see the API Guide for SAP Cloud ALM.

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More

Request and Implement SAP Cloud ALM via SAP for Me

Application help for SAP Cloud ALM for Operations

SAP Cloud ALM for operations (SAP Support Portal)

Expert Portal for SAP Cloud ALM Operations (SAP Support Portal), with detailed configuration instructions  
Involve Local SAP BTP Tools for Operations^

Navigate from central observability capabilities of SAP Cloud ALM to local SAP BTP expert tools or trigger them from SAP Cloud ALM, such as for a use-case specific root cause analysis or for an automated alert remediation.

There are various options you can use to monitor applications and services on SAP BTP, provided natively by the platform itself. Your choice may depend on the SAP BTP runtime you work in. But let's first look at the runtime-agnostic SAP Job Scheduling service.

#### SAP Job Scheduling Service

With SAP Job Scheduling service, you can define and manage jobs that run once or on a recurring schedule. Use this runtime-agnostic service to schedule action endpoints in your application (Cloud Foundry or Kyma) or long-running processes using Cloud Foundry tasks. Use REST APIs to schedule jobs, including long-running jobs asynchronously, and create multiple schedule formats for simple and complex recurring schedules. Manage jobs and tasks and manage schedules with a web-based user interface. Enable integration with the SAP Alert Notification service that sends events for both failed and successful job executions, ensuring that you stay informed about the status of your jobs and can act upon them. To monitor your job executions in SAP Cloud ALM with the job monitoring integration, mark the job in the Dashboard UI or using the REST API. For more information, see SAP Job Scheduling Service.

#### Monitoring Applications in the Cloud Foundry and Kyma Runtimes

SAP Cloud Logging is an observability service to store, visualize, and analyze application logs, metrics, and traces from SAP BTP Cloud Foundry Kyma, Kubernetes and other runtime environments. It enables application developers and operators to analyze their SAP BTP workloads regarding performance, errors, usage, and other characteristics. For Cloud Foundry and Kyma, it offers an easy integration by providing predefined contents to investigate load, latency, and error rates of the observed applications based on their requests and correlate them with additional data.

In Cloud Foundry, SAP Cloud Logging is the successor of the SAP Application Logging service. In the Cloud Foundry runtime, you can still use the SAP Application Logging service for SAP BTP to access logs from your applications.

For Kyma, find detailed guidance in [Integrate with SAP Cloud Logging](#) and [SAP Cloud Logging: Ingest via Kyma Runtime](#).

#### Monitoring Applications in the Neo Environment

In the Neo environment, you can use the cockpit to monitor applications:

##### Monitoring Applications Using the Cockpit in the Neo Environment

Application Type

Monitoring Options

More Information

Java applications

Application monitoring

Custom thresholds

Availability checks

Custom JMX checks

Heap and thread dumps

Email alert notifications

Monitoring APIs

Application logging

Logs API

Monitoring Java Applications

Using Logs in the Cockpit for Java Applications

SAP HANA XS applications

Application monitoring

Availability checks

Custom HTTP checks

Email alert notifications

Monitoring APIs

Health statistics for SAP HANA database instances

Monitoring Database Systems

HTML5 applications

Application monitoring

Custom checks

Email alert notifications

Monitoring APIs

Application logging

Logs API

Monitoring HTML5 Applications

Using Logs in the Cockpit for HTML5 Applications

You can also connect your Java applications to a Dynatrace SaaS monitoring environment. For more information, see [Agent Activation for Dynatrace](#).

Monitoring Platform and Service Availability

To check the availability status of SAP Cloud services, use the SAP Trust Center.

For planned downtimes, you can also check the general information on Maintenance Windows for SAP Cloud services.

To receive customized notifications regarding planned and unplanned downtimes for your SAP Cloud services and tenants, use Cloud Service Availability Notifications in SAP for Me. There, you can select which type of notifications you would like to receive, and who should be notified based on the affected SAP Cloud service and individual service tenant. Administrator users with an S-user and respective privileges can create subscriptions for other users by email (they do not need an S-user account).

By default, a subscription is created for the user who's named as "IT contact" in the contract.

To configure notifications based on business scenarios that involve multiple services or tenants, you can create bundles.

Alerting

In addition to the general notifications mentioned above, SAP Alert Notification service for SAP BTP lets you know instantly whenever there's an issue with your specific cloud application or its dependencies, independent of the runtime your apps are running in. You can subscribe to events coming from SAP BTP services, hyperscalers, third-party tools such as Dynatrace, or you can create custom alerts for your apps that can be pushed directly into the public APIs of SAP Alert Notification service for SAP BTP.

Furthermore, you can subscribe to notifications of your interest and choose the most appropriate channel for delivery, such as email, instant messages to your corporate chat, a ticketing system (like Jira, ServiceNow, etc.), or SAP Cloud ALM.

For more information, see [SAP Cloud ALM and SAP Alert Notification Service](#).

Ops Automation

To simplify and automate complex manual technical processes that are required to run your applications on SAP BTP, consider using Automation Pilot.

Automation Pilot provides catalogs of commands for application lifecycle management, database management, SAP BTP resource and runtime handling, and many more. These provided out-of-the-box automation commands can be complemented by your own content, such your own scripts, HTTP requests, btp CLI tasks, etc. They can then be used in a low-code/no-code approach to compile recommended actions or complex processes to be executed ad hoc or automatically triggered by SAP Cloud ALM and SAP Alert Notification service services for SAP BTP upon a specific event. This helps you reduce your technical operations efforts and make your scenarios more reliable.

For more information, see [SAP Automation Pilot](#).

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URL: <https://help.sap.com/docs/btp/btp-admin-guide/improve-and-retire>

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Improve and Retire

Once your application has gone live, we recommend that you continually ensure its quality. If you determine that you no longer need it, you should retire it.

#### Migrating from Neo to Multi-Cloud

SAP is integrating more closely with hyperscalers, such as Amazon Web Services, Google Cloud Platform, and Microsoft Azure. If you are currently running scenarios in our Neo environment, please have a look at our migration guide for moving from Neo to the multi-cloud foundation.

#### Migrating from the Neo Environment to the Multi-Cloud Foundation for SAP BTP (Cloud Foundry and Kyma)

#### Retiring Your Application

If your application is no longer needed, the Cloud Administration and the Cloud Development Teams should ensure that it is properly retired (for example, undeploying the application and deleting related service instances and other related platform content such as destinations or role collections), and that all data retention requirements are met.

#### Improving and Maintaining Your Application

Once an application has gone live, the Cloud Development Team is responsible for housekeeping and ongoing improvements.

The team should:

- Regularly verify and test the application to avoid issues caused by library and platform updates, such as a new SAPUI5 version

- Ensure that the application remains compliant with all applicable standards and guidelines

- Fix bugs in a timely manner, and ensure that the user experience remains of high quality and is improved and adapted over time

- Listen to user feedback

- Set up alerting that informs you automatically about issues with your application, for example by using SAP Alert Notification service

- Identify recurring operation tasks and automate them, such as by using Automation Pilot. This service provides catalogs of automated technical operation tasks for handling the lifecycle of apps running on SAP BTP, such as for alert remediation and recurring DevOps tasks.

Adapt/extend your application as required:

For deploying new application versions in the SAP BTP, Cloud Foundry environment, consider using a blue-green deployment, in which you deploy a new application version in parallel to the previous version to test it and to route to the new version only after successful testing and without downtime – for more information, see Multitarget Application Commands for the Cloud Foundry Environment.

In the SAP BTP, Cloud Foundry environment, consider using the Feature Flag service that lets you enable or disable specific features in a production solution, for testing purposes, without having to redeploy or restart the solution.

#### Recommendation

To stay up to date, we recommend that you regularly check the Release Notes (or subscribe to receive automatic notifications) and the SAP Community.

#### Related Information

SAP Alert Notification service

SAP Automation Pilot

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URL: <https://help.sap.com/docs/btp/btp-admin-guide/service-configuration-backups-managed-by-customers>

Title: SAP BTP Service Configuration Backups Managed by Customers | SAP Help Portal

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SAP BTP Service Configuration Backups Managed by Customers

This document aims to clarify the backup strategy for SAP Business Technology Platform (SAP BTP) services, specifically focusing on service configurations managed by you, the customer. It isn't intended to serve as a comprehensive guide for all high availability and disaster recovery topics. For more information on resilience, high availability, and disaster recovery, see Resilience, High Availability, and Disaster Recovery.

Prerequisites

You must be aware which SAP BTP services you're currently using:

If you already have a global account on SAP BTP, contact the Global Account Administrator of the global account for a list of the services that you're currently using. If you're unsure who this person is, contact your SAP Sales representative.

If you don't have a global account yet, see [Getting a Global Account](#).

#### How to Back Up Your Service Configurations

SAP doesn't manage backups of service configurations. You're responsible for backing up your service-specific configurations to ensure you can restore them if they're accidentally deleted. For services with user-specific backup capabilities, it's crucial to regularly back up configurations to prevent data loss and ensure continuity. You must familiarize yourself with the documentation of your specific services to effectively manage and restore configurations when needed.

#### Note

You must back up your service configurations yourself to ensure you have backups. Some services don't include user-specific configurations in Data Backups Managed by SAP.

#### Where to Find Backup Information for User-Specific Configurations

Service configurations vary and can be backed up depending on the specific service you use. If a service allows for configuration backups, backup information is part of your service documentation. In case you're unable to find the information you need, refer to [Getting Support](#) for an overview of available support channels and instructions about how to get assistance.

#### Services Lacking Backup Options for User-Specific Configurations

Certain services don't support the backup of user-specific configurations either due to specifics of the services' functionality (for example, a service that doesn't require any user configurations) or due to this feature being still in development. Check your service's documentation to be aware of the limitations and to plan accordingly. If you need support, refer to [Getting Support](#) for an overview of available support channels and instructions about how to get assistance.

#### Related Information

[Data Backups Managed by SAP](#)

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URL: <https://help.sap.com/docs/btp/comparison-btp-runtimes/target-audience-and-personas>

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URL: <https://help.sap.com/docs/btp/implementing-permanent-solution-for-third-party-cookie-deprecation-final/what-is-purpose-of-this-document>

Title: [What Is The Purpose of This Document?](#) | [SAP Help Portal](#)

[Home](#)[SAP Business Technology Platform \(SAP BTP\)](#)[Implementing a Permanent Solution for the Third-Party Cookie Phas...](#)

[Implementing a Permanent Solution for the Third-Party Cookie Phase-Out](#)[English](#)

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## What Is The Purpose of This Document?

This purpose of this guide is to provide guidance about the potential solutions you can implement regarding limitation of third-party cookie usage.

### Note

Google has revised its strategy (July 22nd, 2024) regarding third-party cookies. Instead of enforcing tracking protection by blocking third-party cookies centrally, they will offer this as a choice to their users across all their web browsing. To allow our users to enable tracking protection, the solutions discussed in this guide remain fully relevant. We updated the guide accordingly and will incorporate future guidance of Google as it becomes available.

### Caution

This guide aims at providing guidance, pointing to the right resources, and sharing our experience about solutions for the limitation of third-party cookie usage. It is not exhaustive, and we give no guarantee for correctness, as the topic is driven by Google and subject to change. Please always refer to the official Google documentation for a complete overview and for latest updates.

Browser manufacturers are presenting users with a choice for tracking protection that limits the usage of third-party cookies. A prominent example is Google Chrome's plan to offer people the possibility to make an informed choice that applies across their web browsing.

To help SAP customers and partners best prepare for the limited usage of third-party cookies, we have come up with a blog series and step-by-step guides to help you set up your solution in an SAP context. Before you read this document, we recommend having a look at the following guides to get familiar with the topic:

Preparing and Testing Your Solution for Third-Party Cookie Phase-Out.

Getting and Applying a Deprecation Trial Token for the Third-Party Cookie Phase-Out.

This document provides information about permanent solutions if your application breaks because of the restriction of third-party cookies when users choose to activate tracking protection. Google has listed multiple options in the Solutions section on its website. At SAP, and in this guide, we focus on two options to preserve third-party cookie use cases: Cookie Having Independent State (CHIPS) (also known as cookie partitioning or partitioned cookies) and Storage Access API (SAA).

While these solutions are also supported by other browsers, there might be some differences in the implementation.

### Note

This guide focuses on Google Chrome only. If you want support for different browsers, make sure to read the browser's official documentation on the topic.

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URL: <https://help.sap.com/docs/btp/preparing-and-testing-your-solution-for-third-party-cookie-deprecation>

Title: What Is The Purpose of This Document? | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)Preparing and Testing Your Solution for Third-Party Cookie Phase-...

Preparing and Testing Your Solution for Third-Party Cookie Phase-OutEnglish

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## What Is The Purpose of This Document?

The focus of this guide is to identify issues related to the phase out of third-party cookies. You will learn how to test your application your extensions and assess whether they are affected. Mitigation strategies are covered in Implementing a Permanent Solution for the Third-Party Cookie Phase-Out guide.

### Note

Google has revised its strategy (July 22nd, 2024) regarding the third-party cookie phase-out. Instead of enforcing tracking protection by blocking third-party cookies centrally, they will offer this as a choice to their users across all their web browsing. To allow our users to enable tracking protection, the solutions discussed in the guide remain fully relevant. We will update the guide accordingly and incorporate any guidance of Google as it becomes available.

### Caution

This guide aims at providing guidance, pointing to the right resources, and sharing our experience on testing the impact of third-party phase-out on applications to developers. It is not exhaustive, and we give no guarantee

for correctness, as the topic is driven by Google and subject to change. Please always refer to the official Google documentation for a complete overview and for latest updates.

Browser manufacturers are presenting users with a choice for tracking protection that limits the usage of third-party cookies. A prominent example is Google Chrome's plan to offer people the possibility to make an informed choice that applies across their web browsing.

While the overall plan and timeline is defined by Google and other vendors, we are striving to make sure that our products are best prepared for our users' choice. Our partners and customers, who have successfully built extensions and applications on top of SAP products, are also working hard to guarantee that their solution will keep working with very minimal impact.

Before you get started, we recommend checking our blog, *Are You Affected by the Third-Party Cookie Phase-Out?*, to get an understanding of the different scenarios. As you might be aware, Google is continuing their 1%-trial for unmanaged users. You have to test differently for this scenario. To that end, the *Test for Breakage* chapter provides further instructions on how to set up the testing environment and how to run the different tests. After you have run these tests, and if you identify a breakage in your application due to the third-party cookie blocking, see the *Auditing Cookie Usage using Chrome DevTools* and *Identifying Cookies* to find tips about identifying the responsible cookies.

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URL: <https://help.sap.com/docs/btp/preparing-and-testing-your-solution-for-third-party-cookie-deprecation/auditing-cookie-usage-using-chrome-devtools>

Title: *Auditing Cookie Usage using Chrome DevTools* | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)Preparing and Testing Your Solution for Third-Party Cookie Phase-...Auditing CookiesAuditing Cookie Usage using Chrome DevTools

Preparing and Testing Your Solution for Third-Party Cookie Phase-OutEnglish

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Auditing Cookie Usage using Chrome DevTools

Auditing cookie usage allows you to identify third-party cookies set by your application.

Make sure to manually go through your entire critical user journey and capture any cookies set or read during this process. For more information, criteria and general guidelines on identifying third-party cookies, see Google's official documentation.

At this stage, we are not yet looking for Google Chrome to block third-party cookies. You can use multiple settings to allow third-party cookies.

Note

Some organizations set policies with regards to cookies and might allow certain domains to set cookies.

This effectively hinders the investigation, as those cookies will not be reported as affected. To properly audit your cookie usage, make sure that you are using Google Chrome version M124 or higher, and check `chrome://policy` for any policies set by your application. If there are any, we recommend using a different Google Chrome instance for your testing, inside a virtual machine for example.

The following tools provided by Chrome help with the analysis:

Chrome DevTools

PSAT Extensions

Chrome DevTools^

The Chrome DevTools are quite powerful and have an overwhelming set of features. To help you during your cookie analysis, please consider the following recommendations.

DevTools Panels

The DevTools panels of interest are:

Console

Network

Application - Identify and inspect third-party cookies

Preserve Logs

If your flow spans across multiple pages or includes redirects, enable the *Preserve log* option to avoid losing console messages or request logs. This option needs to be enabled separately for different panels. For more information, check *Console - Persist messages across page loads* and *Network - Save requests across page loads*.

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URL: <https://help.sap.com/docs/btp/preparing-and-testing-your-solution-for-third-party-cookie-deprecation/identifying-cookies>

Title: Identifying Cookies | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)Preparing and Testing Your Solution for Third-Party Cookie Phase-...Auditing CookiesIdentifying Cookies

Preparing and Testing Your Solution for Third-Party Cookie Phase-OutEnglish

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## Identifying Cookies

While going through your critical user journey, watch the Network and Application tabs of the developer tools for cookie usage. If cookies are third-party, they are likely affected by the tracking protection.

The affected cookies could be set by your application or SAP BTP software. SAP will take action to solve the cookie-related issues for cookies set by SAP BTP. For all other cookies, it is your responsibility to adapt your application.

Both SAP-provided software developed on BTP as well as extension scenarios on BTP use SAP technology and thus work with similar cookies. To distinguish between them – and get a high-level understanding of who is responsible for reacting to the third-party cookie phase-out – consider the hostname that attempts to set or read the cookies.

Below is a table with typical components and examples of their common cookies included with BTP-based software you might encounter during testing, that might be affected by the third-party cookie phase-out. Other cookies that you encounter most likely originate from your application and must be handled by you.

Important are the name of the cookie and the domain. Cookies with the same name can occur for different domains.

ComponentFW67 iption"6öö¶–R W† × ÆP

Application Router

Managed: no issue (same domain enforced)

Standalone: must be configured to partition cookie, if on different domains

@sap/approuter - npm (npmjs.com) !' Additional Cookie Attributes

locationAfterLogin

fragmentAfterLogin

signature

SAP Cloud Identity Services (SCI)

These will be taken care of by SCI using the Storage Access API.

No further action is required with regards to these cookies

Typical host name conventions: \*.accounts.ondemand.com or \*.accounts.cloud.sap but customer's own domain is also possible. In case of own domain, please notice the Use Custom Domain in Identity Authentication details.

JSESSIONID

IDP\_USER

SAP Authorization and Trust Management Service (XSUAA)

Typical host naming convention:

\*.authentication.\*.ondemand.com

\*.authentication.\*.sapcloud.cn

JSESSIONID

X-Uaa-Csrft

Cloud Foundry (CF)

CF Runtime is used as one of the underlying technical platforms for building apps on BTP. Therefore, you may see CF-related cookies, like the session-stickiness cookie.

\_\_VCAP\_ID\_\_

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URL: <https://help.sap.com/docs/btp/sap-btp-neo-environment/regions-and-hosts-available-for-neo-environment>

Title: Regions and Hosts Available for the Neo Environment | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP, Neo EnvironmentRegions in the Neo EnvironmentRegions and Hosts Available for the Neo Environment

SAP BTP, Neo EnvironmentEnglish

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## Regions and Hosts Available for the Neo Environment

Each region represents a geographical location (for example, Europe, US East) where applications, data, or services are hosted.

### Caution

SAP Business Technology Platform, Neo environment will sunset on December 31, 2028, subject to terms of customer or partner contracts.

For more information, see SAP Note 3351844 .

### Tip

This documentation refers to SAP Business Technology Platform, Neo environment. If you are looking for documentation about other environments, see SAP Business Technology Platform .

### Tip

Additional IP outbound addresses will soon be used in all regions in the Neo environment. For more information, see Additional IP Addresses Added to SAP BTP, Neo Runtime Regions. To subscribe to such notifications, see SAP Note 3513325 .

To find out about the regions available for multi-environment subaccounts, see Regions .

## Regions, Hosts, and IP Ranges Available for the Neo Environment

### Region (all operated by SAP)

#### Technical Key

#### Host

#### Cockpit Logon

#### IP Ranges

#### Australia (Sydney)

neo-ap1

ap1.hana.ondemand.com

#### Link

#### Inbound:

157.133.96.0/23

#### Outbound:

130.214.148.64/29, 130.214.148.72/29 and 130.214.140.32/27

#### Brazil (São Paulo)

neo-br1

br1.hana.ondemand.com

#### Link

#### Inbound:

130.214.96.128/25

#### Outbound:

130.214.96.64/29, 130.214.96.72/29 and 130.214.96.96/27

#### Canada (Toronto)

neo-ca1

ca1.hana.ondemand.com

#### Link

#### Inbound:

130.214.174.128/25

#### Outbound:

130.214.174.64/29, 130.214.174.72/29 and 130.214.239.32/28

#### China (Shanghai)

neo-cn1

cn1.platform.sapcloud.cn

#### Link

#### Inbound:

121.91.109.0/24

#### Outbound:

121.91.106.72/29, 121.91.106.64/29 and 130.214.219.0/28

#### Europe (Amsterdam)

neo-eu3

eu3.hana.ondemand.com

Link

Inbound:

130.214.86.0/24 and 130.214.87.0/24

Outbound:

130.214.166.72/29, 130.214.166.64/29 and 130.214.166.80/28

Europe (Frankfurt)

neo-eu2

eu2.hana.ondemand.com

Link

Inbound:

157.133.70.0/24, 157.133.205.0/24, 157.133.204.0/24, and 157.133.206.0/24

Outbound:

130.214.164.64/29, 130.214.190.64/29, 130.214.164.80/29, 130.214.190.80/29, 130.214.253.128/27 and 130.214.48.128/27

Europe (Rot)

neo-eu1

hana.ondemand.com

eu1.hana.ondemand.com

Link

Inbound:

155.56.210.0/24, 155.56.248.0/24, 155.56.228.0/23 and 155.56.227.0/24

Outbound:

130.214.18.64/26, 130.214.160.64/28 and 130.214.160.80/29

Japan (Tokyo)

neo-jp1

jp1.hana.ondemand.com

Link

Inbound:

130.214.112.128/25

Outbound:

130.214.245.40/29, 130.214.245.32/29 and 130.214.245.48/28

Kingdom of Saudi Arabia (Riyadh)

neo-sa1

sa1.hana.ondemand.com

Link

Inbound:

130.214.209.128/25

Outbound:

130.214.223.32/29, 130.214.223.40/29 and 130.214.92.96/27

UAE (Dubai)

neo-ae1

ae1.hana.ondemand.com

Link

Inbound:

130.214.80.128/25

Outbound:

130.214.251.32/29, 130.214.251.40/29 and 130.214.251.48/28

US East (Ashburn)

neo-us1

us1.hana.ondemand.com

Link

Inbound: 130.214.181.0/24

Outbound:

130.214.180.64/29, 130.214.180.72/29 and 130.214.180.96/28

US West (Chandler)

neo-us2

us2.hana.ondemand.com

Link

Inbound:

130.214.128.128/25 and 130.214.129.0/24

Outbound:

130.214.255.40/29, 130.214.255.32/29 and 130.214.119.0/27

US East (Sterling)

neo-us3

us3.hana.ondemand.com

Link

Inbound:

130.214.32.0/24 and 130.214.33.0/24

Outbound:

130.214.178.64/29, 130.214.178.72/29, 130.214.180.88/29, 130.214.180.80/29, 130.214.178.96/27 and 130.214.180.192/27

US West (Colorado Springs)

neo-us4

us4.hana.ondemand.com

Link

Inbound:

130.214.183.0/25

Outbound:

130.214.184.64/29, 130.214.184.72/29 and 130.214.184.80/28

Note

The infrastructure layer for Europe (Rot) runs in SAP Data Centers in Rot and Walldorf.

IP Range Notation<sup>^</sup>

The IP ranges listed here are displayed in the Classless Inter-Domain Routing (CIDR) notation. The CIDR notation is a compact way to specify an IP address and its associated routing prefix. The notation is made up of the IP address, a forward slash character (/), and the number of leading 1-bits in the subnet mask.

For example: The CIDR notation 157.133.246.0/24 consists of the IP ranges between 157.133.246.0 and 157.133.246.255.

For more information about how CIDR ranges represent multiple IP addresses, you can read online about CIDR notation.

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/16a6f68b98074b93a4f331e576f2d955.html>

Title: Standard Plan for Cloud Foundry Runtime (Consumption-Based) | SAP Help Portal

HomeSAP BTP, Cloud Foundry RuntimeSAP BTP, Cloud Foundry Runtime...Standard Plan for Cloud Foundry RuntimeStandard Plan for Cloud Foundry Runtime (Consumption-Based)

SAP BTP, Cloud Foundry RuntimeEnglish

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Standard Plan for Cloud Foundry Runtime (Consumption-Based)

Learn how to set up the standard plan for SAP BTP, Cloud Foundry runtime in the consumption-based commercial model.

Overview<sup>^</sup>

Setting up the standard plan for SAP BTP, Cloud Foundry runtime in the consumption-based model involves one mandatory task and one optional task:

Mandatory: A subaccount administrator enables the Cloud Foundry environment with the standard plan, creating a Cloud Foundry org. For more information about the procedure, see Enabling Standard Plan for Cloud Foundry Runtime (Consumption-Based).

Caution

In the consumption-based commercial model, a default runtime memory quota of 200 GB that can be used with the plan standard is automatically assigned to every subaccount. This doesn't mean that you can use the runtime memory for free, as you're going to be billed based on the actual usage. In the consumption-based commercial model, the quota represents a limit on how much runtime memory all the applications in the Cloud Foundry org can use at any given time.

A global account administrator can change the default quota as described in (optional) Adjusting Runtime Memory Quota for Cloud Foundry Runtime (Consumption-Based).

Optional: A global account administrator can change the default runtime memory quota of 200 GB using the service Cloud Foundry Runtime (APPLICATION\_RUNTIME) and its service plan MEMORY. For more information about the procedure, see (optional) Adjusting Runtime Memory Quota for Cloud Foundry Runtime (Consumption-Based).

These tasks are independent of one another and can be done in any order.

Related Information^

What Is the Consumption-Based Commercial Model?

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/38ecf59cdda64150a102cfaa62d5faab.html>

Title: Basic Platform Concepts | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP BTP Administrator's GuideBasic Platform Concepts

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Basic Platform Concepts

SAP BTP offers users the ability to turn data into business value, compose end-to-end business processes, and build and extend SAP applications quickly.

SAP Business Technology Platform is built on a multi-cloud foundation, which lets you choose from different infrastructures and runtimes. The services and solutions of SAP BTP are available on multiple cloud infrastructure providers, and it supports different runtimes, such as Cloud Foundry, ABAP, and Kyma, as well as multiple regions, and a broad choice of programming languages.

The central point of entry to the platform is the SAP BTP cockpit, where you can access your accounts and applications and manage all activities associated with them.

Check out the SAP Discovery Center to get an overview on the available SAP BTP services, discover missions that guide you step-by-step, and explore reference architectures.

Overview of Global Accounts, Directories, and Subaccounts^

The following diagram depicts the relationship between a global account, its directories, subaccounts, runtime environments, regions, entitlements, and quotas.

Administrative Tasks^

Level

Administrative Tasks

Global Account

Global accounts are region- and runtime-independent.

Usually, for each commercial model (licence type), you get a separate global account. It is however possible to combine two commercial models in one global account. See Commercial Models.

Appoint at least one person as administrator. The administrator is responsible for adding new subaccounts, adding members to a global account, and managing the entitlements.

You purchase entitlements for each global account (according to your commercial model). The administrator of the global account distributes quotas to the individual subaccounts.

Directory (optional)

Directories are groups of subaccounts that you can manage, operate, and analyze together.

Appoint at least one person as administrator. The administrator is responsible for adding new subaccounts, managing members, and managing entitlements.

Subaccount

Each subaccount runs in exactly one region. Within a subaccount, you can enable multiple, different runtime environments.

Appoint at least one person as administrator. The administrator is responsible for adding new members to the subaccount and assigning their business roles.

## Recommendation

For each administrator, we recommend that you also appoint at least one substitute administrator. If the main administrator leaves the company or is unavailable, it's important that you have someone who's available to take over these tasks.

For more information, see [Account Model](#) and [Setting Up Your Account Model](#).

## Capabilities and Services

SAP BTP offers fast in-memory processing, sustainable, agile solutions and services to integrate data and extend applications, and fully embedded analytics and intelligent technologies.

Services enable, facilitate, or accelerate the development of business applications and other platform services on SAP BTP.

For a complete list of services and capabilities, see [Services](#).

For more information, see [Solutions and Services](#).

## Entitlements and Quotas

When you purchase an enterprise account, you're entitled to use a specific set of resources, such as the amount of memory that can be allocated to your applications.

On SAP BTP, all external dependencies such as databases, messaging systems, files systems, and so on, are services. In this context, multitenant applications and environments are considered services.

Each service has one or more service plans available. A service plan is the representation of the costs and benefits for a given variant of a particular service. For instance, a database may be configured with various "T-shirt sizes", each of which is a different service plan.

An entitlement is your right to provision and consume a resource. In other words, entitlements are the service plans that you're entitled to use.

A quota represents the numeric quantity of a service plan that you're entitled to consume in your global account and its subaccounts.

Entitlements are either managed at global account or directory level and are distributed from there to subaccounts.

For more information, see [Entitlements and Quotas](#) and [Managing Entitlements and Quotas Using the Cockpit](#).

## User and Member Management

On SAP BTP, user management takes place at all levels from global account to environment. There are different types of users, such as depending on their roles in the company.

User accounts enable users to log on to SAP BTP, access subaccounts, and to use applications according to the permissions granted to them. We distinguish between two types of users:

Platform users are usually developers, administrators or operators who deploy, administer, and troubleshoot accounts, applications and services on SAP BTP.

Business users use the applications that are deployed to SAP BTP. For example, the end users of SaaS apps or services, such as SAP Build Work Zone, or end users of your custom applications are business users.

Member management refers to managing permissions for platform users. A member is a user who is assigned to an SAP BTP global account or subaccount. Administrators can add users to global accounts and subaccounts and assign roles to them as needed. You can use predefined roles, for example the administrator role for managing subaccount members.

User management refers to managing authentication and authorization for your business users.

For more information, see [User and Member Management](#).

## Account Model

The SAP BTP cockpit is structured according to global accounts and subaccounts. You can also use directories and labels to organize your account model.

### Global Accounts

A global account is the realization of a contract you or your company has made with SAP.

A global account is used to manage subaccounts, members, entitlements and quotas. You receive entitlements and quotas to use platform resources per global account and then distribute the entitlements and quotas to the subaccount for actual consumption. There are two types of commercial models for global accounts: consumption-based model and subscription-based model. See [Commercial Models](#).

Global accounts are region- and environment-independent. Within a global account, you manage all of your subaccounts, which in turn are specific to one region.

### Subaccounts

In subaccounts, you deploy applications, use services, and manage your subscriptions. You can use them to manage members, authorizations, and entitlements.

A global account can contain one or more subaccounts in which you deploy applications, use services, and manage your subscriptions. Subaccounts in a global account are independent from each other. This is important to consider with respect to security, member management, data management, data migration, integration, and so on, when you plan your landscape and overall architecture.

Each subaccount is associated with a region, which is the physical location where applications, data, or services are hosted. The specific region is relevant when you deploy applications and access the SAP BTP cockpit using the corresponding cockpit URL. The region assigned to your subaccount doesn't have to be directly related to your location. You could be located in the United States, for example, but operate your subaccount in Europe.

The entitlements and quotas that have been purchased for a global account have to be assigned to the individual subaccounts.

Global accounts and subaccounts are completely independent of user accounts. For more information, see [User and Member Management](#).

#### Subaccounts, Cloud Foundry Orgs and Kyma Clusters

When you enable the Cloud Foundry environment in one of your subaccounts, the system automatically creates a Cloud Foundry org for you. The subaccount and the org have a 1:1 relationship and the same navigation level in the cockpit (even though they may have different names). You can create spaces within that Cloud Foundry org. Spaces let you further break down your account model and use services and functions in the Cloud Foundry environment.

For more information about Cloud Foundry orgs and spaces, see the Cloud Foundry documentation at <https://docs.cloudfoundry.org/concepts/roles.html>.

When you enable the Kyma environment in one of your subaccounts, you can provision one or more Kubernetes clusters equipped with the latest version of the open-source project "Kyma". The subaccount and the Kyma cluster have a 1:n relationship - you can choose to create more than one cluster right away, or add clusters later.

Within each Kyma cluster, you can create namespaces to organize your workloads and control access to resources. Namespaces let you further break down your account model and use services and functions in the Kyma environment. For more information, see the Kubernetes documentation at <https://kubernetes.io/docs/concepts/overview/working-with-objects/namespaces/>.

#### Directories

With directories, you can organize and manage your subaccounts according to your technical and business needs.

A directory can contain directories and subaccounts to create a hierarchy. Using directories to group other directories and subaccounts is optional - you can still create subaccounts directly under your global account. You can create a hierarchical structure that is up to 7 levels deep. The highest level of a given path is always the global account and the lowest is a subaccount, which means that you can have up to 5 levels of directories between the global account and the lowest level subaccount.

Directories enable you to:

- Group and filter directories and subaccounts

- Monitor usage and costs for contracts that use the consumption-based commercial model

Optionally, you can also enable the following features in your directories:

**Manage Entitlements:** Enables the assignment of a quota for services and applications to the directory from the global account quota, for distribution to the directory's subaccounts.

When you assign entitlements to a directory, you express the entitlements and maximum quota that can be distributed across its children subaccounts. You also have the option to choose the auto-assignment of a set amount of quota to all subaccounts created or moved to that directory. Subaccounts that are already in the directory when you select that option will not be auto-assigned quota.

#### Note

If you've enabled the Manage Entitlements feature for a given directory, you must first assign the necessary entitlements and maximum allowed quota from the global account to that directory. Then you can distribute this "reserved" quota to any of the directory's child subaccounts.

**Manage Authorizations:** Enables authorization management for the directory. For example, it allows certain

users to manage directory entitlements. You can only use this feature in combination with the Manage Entitlements feature.

#### Labels

Labels are user-defined words or phrases that you can assign to various entities in SAP BTP to categorize them in your global account, to identify them more easily. They let you organize and filter your directories, subaccounts, instances and subscriptions within your global account.

For more information, see Labels.

#### Related Information

##### Account Model

Managing Global Accounts Using the Cockpit

Managing Directories Using the Cockpit

Managing Subaccounts Using the Cockpit

Setting Up Your Account Model

##### Commercial Models

SAP BTP offers two different commercial models for enterprise accounts.

**Consumption-based commercial model:** Your organization receives access to all current and future services that are eligible for this model. You have complete flexibility to turn services on and off and to switch between services as your business requires throughout the duration of your contract. This commercial model is available in the following flavors: SAP BTP Enterprise Agreement (SAP BTPEA), Cloud Platform Enterprise Agreement (CPEA), and Pay-As-You-Go for SAP BTP.

For more information, see [What Is the Consumption-Based Commercial Model?](#)

**Subscription-based commercial model:** Your organization subscribes only to the services that you plan to use. You can then use these services at a fixed cost, irrespective of consumption.

For more information, see [What Is the Subscription-Based Commercial Model?](#)

#### Note

You can use both commercial models, either in separate global accounts or in the same global account depending on your business needs. Contact your SAP account executive or sales representative for more information. Note however that you cannot mix more than one flavor of the consumption-based commercial model in the same global account.

For information about service availability, prices, and estimators, see <https://www.sap.com/products/technology-platform/solutions.html> and <https://www.sap.com/products/technology-platform/integration-suite/pricing.html>.

You can also view the service catalog via the SAP Discovery Center.

##### Environments

Environments constitute the actual platform-as-a-service offering of SAP BTP that allows for the development and administration of business applications. Environments are anchored in SAP BTP on subaccount level. Each environment comes equipped with specific tools, technologies, and runtimes that you need to build applications. So a multi-environment subaccount is your single address to host a variety of applications and offer diverse development options. One advantage of using different environments in one subaccount is that you only need to manage users, authorizations, and entitlements once per subaccount, and thus grant more flexibility to your developers.

You can use different environments within the same subaccount: Choose multi-environment subaccounts to use Cloud Foundry, ABAP, and Kyma within the same subaccount. However, inside such a multi-environment subaccount, you cannot use Neo. To use Neo, you create a separate Neo subaccount. Neo cannot be combined with other environments in the same subaccount.

SAP offers the following environments:

##### Cloud Foundry Environment

The Cloud Foundry environment enables you to develop new business applications and business services, supporting multiple runtimes, programming languages, libraries, and services. You can leverage a multitude of buildpacks, including community innovations and self-developed buildpacks.

##### ABAP Environment

The ABAP environment allows you to create extensions for ABAP-based products, such as SAP S/4HANA Cloud, and develop new cloud applications. You can transform existing ABAP-based custom code or extensions to the cloud.

To plan and set up your landscape and lifecycle management in the ABAP environment, see [ABAP Lifecycle](#)

Management.

#### Kyma Environment

SAP BTP, Kyma runtime provides a fully managed cloud-native Kubernetes application runtime based on the open-source project "Kyma". Based on modular building blocks, Kyma runtime includes all the necessary capabilities to simplify the development and to run enterprise-grade cloud-native applications.

#### Neo Environment

The Neo environment lets you develop HTML5, Java, and SAP HANA extended application services (SAP HANA XS) applications. You can also use the UI Development Toolkit for HTML5 (SAPUI5) to develop rich user interfaces for modern web-based business applications.

If you consider creating Neo subaccounts, see [Migrating from the Neo Environment to the Multi-Cloud Foundation for SAP BTP \(Cloud Foundry and Kyma\)](#) to learn about the Neo environment and the multi-cloud foundation.

#### Environment Instances

To actually use an environment in a subaccount, you must enable it, which creates an instance of that environment. There are several ways to create environment instances:

In the SAP BTP cockpit, on the subaccount overview, choose **Enable**.

In the SAP BTP cockpit, under **Service Marketplace**. Here, you get more information, such as the available plans and links to further information.

Using the `bt` CLI command `bt create accounts/environment-instance`

#### Related Information

[SAP BTP, Serverless Runtime](#)

[Consuming SAP BTP Services from Various Environments](#)

#### Regions

You can deploy applications in different regions. Each region represents a geographical location (for example, Europe, US East) where applications, data, or services are hosted.

Regions are provided either by SAP or by our Infrastructure-as-a-Service (IaaS) partners Amazon Web Services (AWS), Microsoft Azure, Google Cloud, and Alibaba Cloud. The third-party region providers operate the infrastructure layer of the regions, whereas SAP operates the platform layer and Cloud Foundry.

A region is chosen at the subaccount level. For each subaccount, you select exactly one region (that is one data center).

For more information, see [Regions](#).

#### Selecting a Region

When deciding on the location of your Platform as a Service (PaaS), consider existing Software as a Service (SaaS) and Infrastructure as a Service (IaaS) and try to locate it close to those or even in the same data center.

You can also optimize application performance (response time, latency) by selecting a region that's geographically close to your users. However, the selection of a region also depends on many other factors:

Check the availability of specific services in the individual regions.

Ensure that you comply with security requirements, such as country- or industry-specific data privacy regulations.

Consider the location of other cloud offerings you're using. You might have to locate your solutions in the same data center.

For a complete overview of the availability of services in the different regions, see [Services](#).

#### Note

To work with the Kyma environment, you need to specify the region for both your subaccount and the cluster.

Note that there is a number of regions available within each plan. For details, see [Regions for the Kyma Environment](#).

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/45341f37cf6e4738a4b7cd20f18350de.html>

Title: [What Is Application Autoscaler?](#) | SAP Help Portal

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Application AutoscalerEnglish

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What Is Application Autoscaler?

Automatically scale your applications to meet their dynamic resource needs.

Application Autoscaler lets you automatically increase or decrease the number of your application instances based on the policies you have defined.

#### Tip

The English version of this guide is open for contributions and feedback using GitHub. This allows you to get in contact with responsible authors of SAP Help Portal pages and the development team to discuss documentation-related issues. To contribute to this guide, or to provide feedback, choose the corresponding option on SAP Help Portal:

**Feedback Create issue:** Provide feedback about a documentation page. This option opens an issue on GitHub.

**Feedback Edit page:** Contribute to a documentation page. This option opens a pull request on GitHub.

You need a GitHub account to use these options.

More information:

[Contribution Guidelines](#)

[Introduction Video](#)

[Introduction Blog Post](#)

[Features](#)

[Scale applications on CF automatically](#)

Automatically increase or decrease the number of application instances to meet the changing demands of application resources.

[Make your scaling resource-specific](#)

Scale your applications on the basis of any standard metric type, such as memory consumed, CPU, response time, or throughput.

[Define custom metrics](#)

Scale your applications by defining custom metrics that match their requirements.

[Manage your costs effectively](#)

Only consume and pay for the compute resources, which your applications require.

[Environment](#)

This service runs in the Cloud Foundry environment.

[Regional Availability](#)

For information on the regional availability of Application Autoscaler, see SAP Discovery Center.

For more information, see [Regions](#).

[Documentation Feedback Survey](#)

[Give Feedback](#)

Help us improve our documentation. Please take this two-minute survey and let us know what you think.

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Title: Basic Platform Concepts | SAP Help Portal

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Basic Platform Concepts

SAP BTP offers users the ability to turn data into business value, compose end-to-end business processes, and build and extend SAP applications quickly.

SAP Business Technology Platform is built on a multi-cloud foundation, which lets you choose from different infrastructures and runtimes. The services and solutions of SAP BTP are available on multiple cloud infrastructure providers, and it supports different runtimes, such as Cloud Foundry, ABAP, and Kyma, as well as multiple regions, and a broad choice of programming languages.

The central point of entry to the platform is the SAP BTP cockpit, where you can access your accounts and applications and manage all activities associated with them.

Relationship between SAP BTP Accounts and Associated Activities

Solutions and Servicesà^

SAP BTP offers fast in-memory processing, sustainable, agile solutions and services to integrate data and extend applications, and fully embedded analytics and intelligent technologies.

Services enable, facilitate, or accelerate the development of business applications and other platform services on SAP BTP.

For a complete list of services and capabilities, see SAP Discovery Center: Services.

For more information, see Solutions and Services.

Regionsà^

You can deploy applications in different regions. Each region represents a geographical location (for example, Europe, US East) where applications, data, or services are hosted.

Regions are provided either by SAP or by our Infrastructure-as-a-Service (IaaS) partners Amazon Web Services (AWS), Microsoft Azure, Google Cloud, and Alibaba Cloud. The third-party region providers operate the infrastructure layer of the regions, whereas SAP operates the platform layer and Cloud Foundry.

A region is chosen at the subaccount level. For each subaccount, you select exactly one region (that is one data center).

For more information, see Regions.

Environmentsà^

Environments constitute the actual platform-as-a-service offering of SAP BTP that allows for the development and administration of business applications. Environments are anchored in SAP BTP on subaccount level.

SAP BTP provides the following environments:

Cloud Foundry Environment

The Cloud Foundry environment enables you to develop new business applications and business services, supporting multiple runtimes, programming languages, libraries, and services. You can leverage a multitude of buildpacks, including community innovations and self-developed buildpacks.

ABAP Environment

The ABAP environment allows you to create extensions for ABAP-based products, such as SAP S/4HANA Cloud, and develop new cloud applications. You can transform existing ABAP-based custom code or extensions to the cloud.

Kyma Environment

SAP BTP, Kyma runtime provides a fully managed cloud-native Kubernetes application runtime based on the open-source project "Kyma". Based on modular building blocks, Kyma runtime includes all the necessary capabilities to simplify the development and to run enterprise-grade cloud-native applications.

Neo Environment

The Neo environment lets you develop HTML5, Java, and SAP HANA extended application services (SAP

HANA XS) applications. You can also use the UI Development Toolkit for HTML5 (SAPUI5) to develop rich user interfaces for modern web-based business applications.

#### Caution

SAP Business Technology Platform, Neo environment will sunset on December 31, 2028, subject to terms of customer or partner contracts.

For more information, see SAP Note 3351844 .

#### Enterprise and Trial Accounts

SAP BTP provides different types of global accounts, enterprise and trial. The type you choose determines pricing, conditions of use, resources, available services, and hosts.

A trial account lets you try out the platform for free. Access is open to everyone. Trial accounts are intended for personal exploration, and not for production use or team development. They allow restricted use of the platform resources and services.

For more information, see Trial Accounts and Free Tier.

An enterprise account is usually associated with one SAP customer or partner and contains their purchased entitlements to platform resources and services. It groups together different subaccounts that an administrator makes available to users for deploying applications.

For more information, see Enterprise Accounts.

#### Commercial Models

SAP BTP offers two different commercial models:

Consumption-based commercial model: Your organization receives access to all current and future services that are eligible for this model. You have complete flexibility to turn services on and off and to switch between services as your business requires throughout the duration of your contract. This commercial model is available in the following flavors: SAP BTP Enterprise Agreement (SAP BTPEA), Cloud Platform Enterprise Agreement (CPEA), and Pay-As-You-Go for SAP BTP.

For more information, see What Is the Consumption-Based Commercial Model?

Subscription-based commercial model: Your organization subscribes only to the services that you plan to use. You can then use these services at a fixed cost, irrespective of consumption.

For more information, see What Is the Subscription-Based Commercial Model?

#### Account Model

The SAP BTP cockpit is structured according to global accounts, directories, and subaccounts:

A global account is the realization of a contract you or your company has made with SAP. A global account is used to manage subaccounts, members, entitlements and quotas. You receive entitlements and quotas to use platform resources per global account and then distribute the entitlements and quotas to the subaccount for actual consumption. There are two types of commercial models for global accounts: consumption-based model and subscription-based model. See Commercial Models

With directories, you can organize and manage your subaccounts according to your technical and business needs.

In subaccounts, you deploy applications, use services, and manage your subscriptions. You can use them to manage members, authorizations, and entitlements.

In the SAP BTP, Cloud Foundry environment, the subaccount is divided into one or more spaces, which is where application development, deployment, and maintenance take place.

For more information, see Account Model.

#### Entitlements and Quotas

When you purchase an enterprise account, you're entitled to use a specific set of resources, such as the amount of memory that can be allocated to your applications.

On SAP BTP, all external dependencies such as databases, messaging systems, files systems, and so on, are services. In this context, multitenant applications and environments are considered services.

Each service has one or more service plans available. A service plan is the representation of the costs and benefits for a given variant of a particular service. For instance, a database may be configured with various "T-shirt sizes", each of which is a different service plan.

An entitlement is your right to provision and consume a resource. In other words, entitlements are the service plans that you're entitled to use.

A quota represents the numeric quantity of a service plan that you're entitled to consume in your global account and its subaccounts.

For more information, see [Entitlements and Quotas](#).

#### User and Member Management<sup>^</sup>

On SAP BTP, user management takes place at all levels from global account to environment. There are different types of users, such as depending on their roles in the company.

User accounts enable users to log on to SAP BTP, access subaccounts, and to use applications according to the permissions granted to them. We distinguish between two types of users:

Platform users are usually developers, administrators or operators who deploy, administer, and troubleshoot accounts, applications and services on SAP BTP.

Business users use the applications that are deployed to SAP BTP. For example, the end users of SaaS apps or services, such as SAP Build Work Zone, or end users of your custom applications are business users.

Member management refers to managing permissions for platform users. A member is a user who is assigned to an SAP BTP global account or subaccount. Administrators can add users to global accounts and subaccounts and assign roles to them as needed. You can use predefined roles, for example the administrator role for managing subaccount members.

User management refers to managing authentication and authorization for your business users.

For more information, see [User and Member Management](#).

#### Tools, Programming Models, Programming Languages, and APIs<sup>^</sup>

SAP BTP provides various programming languages and tools for your development project:

SAP BTP includes many tools to help you develop and manage applications, and connect them to your on-premise systems. For more information, see [Tools](#).

SAP BTP supports many different programming languages; the availability of each depends on the development environment you're using. For more information, see [Programming Languages](#).

The SAP Cloud Application Programming Model offers a consistent end-to-end programming model that includes languages, libraries, and APIs that are tailored for full-stack development on SAP BTP. For more information, see [Programming Models](#).

Depending on your use case, you can choose between different offerings for continuous integration and delivery. For more information, see [Continuous Integration and Delivery \(CI/CD\)](#).

Discover and consume APIs to manage, build, and extend the core capabilities of SAP BTP. For more information, see [APIs](#).

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/btp-getting-started>

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#### Getting Started

Once you're familiar with the basic concepts of SAP BTP, you can start your first workflows or check out further resources that help you get started quickly.

#### Onboarding Guides<sup>^</sup>

The SAP BTP Administrator's Guide helps you plan your development projects on SAP BTP from onboarding to SAP Cloud Identity Services, through setting up the correct organizational structure, to creating an account and security model, to developing and operating applications.

The SAP BTP Developer's Guide helps you define the correct methodologies and tools for your development project.

#### Getting Started Workflows<sup>^</sup>

Learn how to get a global account and get started with a trial or an enterprise account on SAP BTP. A trial account lets you try out SAP BTP for free, and within an enterprise account, you can use so-called free tier service plans for free. See [Trial Accounts and Free Tier](#).

Depending on your use case, follow the appropriate workflow:

#### Getting a Global Account

[Getting Started in the Cloud Foundry Environment](#)

[Getting Started in the ABAP Environment](#)

[Getting Started in the Kyma Environment](#)

#### Additional Resources<sup>^</sup>

The SAP Developer Center provides further resources to get you started:

[SAP Discovery Center](#)

[SAP BTP in SAP Developer Center](#)

[Start Developing on SAP BTP Tutorials](#)

[Related Information](#)

[SAP BTP Administrator's Guide](#)

[SAP BTP Developer's Guide](#)

[Access the Cockpit](#)

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[Getting Support](#)

To get help, use the available support channels provided by SAP for Me.

Note

This section is applicable for all multicloud environments. To find which environments are available in your region, see [Regions](#).

[Prerequisites](#)

Before you report a case, ensure the following:

[Check Platform Status](#)

Running on the AWS, Azure, or GCP regions: visit [SAP Trust Center](#). You can check:

availability by service on [Cloud Status](#) tab page;

availability by region on the [Data Center](#) tab page.

Running on the China (Shanghai) region: check [China \(Shanghai\) region status](#) page.

Running on the Government Cloud (US) region: planned downtimes and outage communication are sent through e-mail to the initial administrator of your global account.

[Check Tools Versions](#)

Make sure that you have the latest versions of the tools you're using (recommended), or at least the versions you're using are still supported. For more information, see [SAP Development Tools](#).

Note

Trial users can ask for support by posting a question in [SAP Community](#). However, they should first check for answers in [SAP Community](#).

[Try Out Built-In Support Within SAP BTP Cockpit](#)

When using SAP BTP cockpit as an S-user, try Built-In Support to get customer help within the cockpit. See [Built-In Support](#).

[Learn the Best Practices and Offerings](#)

Familiarize with the guidelines how to engage with support. See [Support from SAP Best Practices and Offerings](#).

[Procedure](#)

To report a case in SAP for Me, proceed as follows:

Open SAP for Me and log on.

Open the [Services & Support](#) dashboard.

(Optional) You can use the [Knowledge Search](#) tab for existing solutions in:

[SAP Notes & KBAs](#)

[SAP Community](#)

[SAP Support Portal](#)

[Product Documentation](#)

Use the [Get Support](#) application within SAP for Me to get assistance via the available support channels.

For more information about the specifics of each support channel and to learn how to report technical issues, see [Get Support Application](#).

For any other non-technical assistance, see [Customer Interaction Center](#).

[Creating a Support User](#)

Sometimes you might need to create a support user to troubleshoot issues in your system. For more information about how and when to create a support user, see 3543065 - Best Practices for Creating Support Users in Your BTP System.

Related Information

Users and Authorizations for SAP One Support Launchpad

Gather Support Information

Platform Updates and Notifications

Cloud Availability Center

Road Map for Cloud Foundry Environment

Road Map for Kyma Runtime

Road Map for ABAP Environment

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/business-users-2e684942eb964bfa435b5ed6702824a>

Title: Business Users | SAP Help Portal

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Business Users

Business users use the applications that are deployed to SAP BTP. For example, the end users of SaaS apps or services, such as SAP Build Work Zone, or end users of your custom applications are business users.

Application developers (platform users) create and deploy application-specific security artifacts for business users, such as scopes. Administrators use these artifacts to assign roles, build role collections, and assign these role collections to business users or user groups. In this way, they control the users' permissions in the application.

For business users, there's a default identity provider, too. We expect that you have your own identity provider.

Recommendation

We strongly recommend that you configure your custom tenant of SAP Cloud Identity Services as the identity provider and connect SAP Cloud Identity Services to your own corporate identity provider.

For more information, see Trust and Federation with Identity Providers.

User Managementà^

User management refers to managing authentication and authorization for your business users.

To manage your business users:

Configure trust to an application identity provider in your subaccount.

Create shadow users in your subaccount for your business users in your identity provider.

When a user accesses a resource, SAP BTP redirects the user to the identity provider for authentication. You assign authorizations to shadow users in SAP BTP.

Assign role collections either directly to users or map them to user groups.

The role collections were either delivered from the applications to which you subscribed or custom developed by your team.

To learn more about user management, see Security Administration: Managing Authentication and Authorization.

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/change-configuration-settings>

Title: Change Configuration Settings | SAP Help Portal

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Change Configuration Settings

Change the configuration settings to customize the behavior of the btp CLI.

Contextà^

There are a few configuration settings that you can set to customize how the btp CLI works.

These settings are saved to the configuration file, which governs how the btp CLI works. You can have more than one configuration file for working in different accounts at the same time by logging in with the `--config PATH` option and then also providing it with each command call. See [Specify the Location of the Configuration File](#).

To find out where the configuration file is stored, use `btp --info`.

#### Procedure

To change a setting, run:

```
btp [OPTIONS] set config [--format FORMAT] [--verbose BOOL] [--login.sso MODE] [--login.securestore BOOL]
[--login.showglobalaccounts BOOL] [--target.hierarchy BOOL] [--theme THEME]
```

#### Parameter

##### Description

`--format FORMAT`

The output format of commands. Valid values: text (default), json.

`--verbose BOOL`

If set to true, command output includes tracing information for support. Valid values: false (default), true.

`--login.sso MODE`

Single sign-on mode of btp login. Valid values: none (default), browser, manual. For example, if you want btp login to always open a browser for login at your identity provider, use `btp set config --login.sso browser`.

`--login.securestore BOOL`

If true, your login session will be stored in your OS secure store (only macOS/Windows).

`--login.showglobalaccounts BOOL`

If set to true, btp login will always prompt you to choose a global account.

`--target.hierarchy BOOL`

If set to true, btp target displays the full hierarchy of all global accounts. Valid values: false (default), true

See [Set a Target for Subsequent Commands with btp target](#).

`--theme THEME`

Theme setting for the CLI. Use environment variable `NO_COLOR=true` to force the theme to 'none', regardless of this setting.

To list the current settings, run:

```
btp [OPTIONS] list config
```

To reset individual or all settings back to their default values, run:

```
btp [OPTIONS] reset config [--format] [--verbose] [--login.sso] [--login.securestore] [--login.showglobalaccounts]
[--target.hierarchy] [--theme] [--all]
```

#### Related Information

[Specify the Location of the Configuration File](#)

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/change-output-format-to-json>

Title: [Change the Output Format to JSON | SAP Help Portal](#)

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Change the Output Format to JSON

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Change the Output Format to JSON

Use the `--format json` option to change the output format to JSON. This is the recommended output format for automation with the btp CLI.

#### Context

The standard output format of the btp CLI is plain text, formatted in a way that an interactive user of the btp CLI can easily read it and therefore often limited to the most relevant information. We don't recommend using this plain text output for automatic parsing and extracting information. The text output is not stable enough, as, for example, column width or the appearance of new lines may depend on the actual content of a response.

If you're using responses of the btp CLI for automation, such as in scripts, we recommend the JSON output format, as JSON is a data format that can be parsed by most programming languages. The JSON output is much more stable than the plain text output and it often includes more information in the JSON response than

the respective text output. Because it is optimized for post-processing, it doesn't include non-essential information such as an OK at the end of a successful call.

#### Tip

To set the command output to json persistently, you can change the configuration settings with `btpp set config --format json`. This makes using the `--format json` option obsolete. See [Change Configuration Settings](#). See this [Getting BTP resource GUIDs with the btpp CLI](#) blog post for some examples.

#### Procedure

Start commands with `--format json`. Currently, the only valid value is `json`.

Example

~

```
btpp --format json list accounts/subaccount
```

Results

Example with text output

Example

~

```
btpp list accounts/subaccount
```

subaccounts in global account c8XXXX3d-6XX7-4XXb-8dXX-89XXXX372eXX...

subaccount id:

display name:

subdomain:

region:

beta-enabled:

parent id:

state:

state message:

88XXXX80-9844-4XX7-8XXd-beXXXX42bfb2

my-subaccount-1

my-CF-subdomain1

us10-TEST

false

c8XXXX3d-6XX7-4XXb-8dXX-89XXXX372eXX

OK

Subaccount created.

ebXXXX3e-f3a2-4XXf-a080-3eXXXX9b9ced

my-subaccount-2

my-CF-subdomain2

us10-TEST

false

c8XXXX3d-6XX7-4XXb-8dXX-89XXXX372eXX

OK

Subaccount created.

Example

Example with JSON output

Example

~

```
btpp --format json list accounts/subaccount
```

```
{
```

```
"value": [
```

```
{
```

```
"guid": "88XXXX80-9844-4XX7-8XXd-beXXXX42bfb2",
```

```
"displayName": "my-subaccount-1",
```

```
"globalAccountGUID": "c8XXXX3d-6XX7-4XXb-8dXX-89XXXX372eXX",
```

```
"parentGUID": "c8XXXX3d-6XX7-4XXb-8dXX-89XXXX372eXX",
```

```
"parentType": "ROOT",
```

```
"region": "us10-TEST",
"subdomain": "my-CF-subdomain1",
"betaEnabled": false,
"usedForProduction": "UNSET",
"description": null,
"expiryDate": null,
"state": "OK",
"stateMessage": "Subaccount created.",
"createdDate": 1602759737909,
"createdBy": "name@email.com",
"modifiedDate": 1602759737909,
"zoneId": null
},
{
"guid": "ebXXXX3e-f3a2-4XXf-a080-3eXXXX9b9ced",
"displayName": "CF TEST",
"globalAccountGUID": "c8XXXX3d-6XX7-4XXb-8dXX-89XXXX372eXX",
"parentGUID": "c8XXXX3d-6XX7-4XXb-8dXX-89XXXX372eXX",
"parentType": "ROOT",
"region": "us10-TEST",
"subdomain": "my-CF-subdomain2",
"betaEnabled": false,
"usedForProduction": "UNSET",
"description": null,
"expiryDate": null,
"state": "OK",
"stateMessage": "Subaccount created.",
"createdDate": 1602760244477,
"createdBy": "name@email.com",
"modifiedDate": 1602760244477,
"zoneId": null
}
]
}
```

Related Information

Change Configuration Settings

---

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/configuring-serverless>

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/creating-service-keys>

Title: *Creating Service Keys | SAP Help Portal*

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### Creating Service Keys

You can use service keys to generate credentials to communicate directly with a service instance. Once you configure them for your service, local clients, apps in other spaces, or entities outside your deployment can access your service with these keys.

You can use the SAP BTP cockpit or the Cloud Foundry Command Line Interface to create service keys:

Create Service Keys Using the Cockpit

Create Service Keys Using the Cloud Foundry Command Line Interface

For more information on service keys, see <https://docs.cloudfoundry.org/devguide/services/service-keys.html>.

---

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/creating-user-provided-service-instances>

Title: *Creating User-Provided Service Instances | SAP Help Portal*

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### Creating User-Provided Service Instances

User-provided service instances enable you to use services that are not available in the marketplace with your applications running in the Cloud Foundry environment.

You can create user-provided service instances using the SAP BTP cockpit or the Cloud Foundry Command Line Interface:

Create User-Provided Service Instances Using the Cockpit

Create User-Provided Service Instances Using the Cloud Foundry Command Line Interface

For more information on user-provided service instances, see <https://docs.cloudfoundry.org/devguide/services/user-provided.html>.

---

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/custom-code-analysis-for-sap-btp-configuration-guide>

Title: *Performing a Custom Code Analysis | SAP Help Portal*

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### Performing a Custom Code Analysis

This guide describes how to perform remote custom code analysis using the Analyze Custom Code app.

The Analyze Custom Code app in the SAP BTP ABAP environment can act as the ABAP Test Cockpit (ATC), a central check system analyzing the ABAP source code located in your on-premise system. It enables you to identify functional or security vulnerabilities in your ABAP source code. This includes security checks performed by the SAP Code Vulnerability Analyzer (CVA). Further cloud use cases for ATC in the Analyze Custom Code app involve the migration to S/4HANA, performance checks, and ABAP Cloud readiness checks.

In the following chapters, you will learn how to create a custom code analysis project, analyze the findings, and, in addition, perform dependency and complexity analyses.

Related Informationà

Code Vulnerability Analyzer Checks

---

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/enable-command-autocompletion>

Title: SAP Help Portal | SAP Online Help

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Enable Command Autocompletion

Use command autocompletion to save keystrokes when entering command actions, group-object combinations, and their parameters in the SAP BTP command line interface (btp CLI).

Contextà^

Autocompletion in btp CLI currently supports the following shells:

Bash

PowerShell

Zsh

Note

The respective shell must be installed on your operating system before enabling autocomplete. Additionally, the btp CLI executable must be available on the path for autocompletion to initialize. To install it on the path, follow the steps outlined inDownload and Start Using the btp CLI Client.

Once autocomplete is enabled (it's disabled by default), you use the autocomplete feature as follows in the command line:

Enter a partial command action, group-object combination, or parameter, and then press the Tab key. The command line either automatically completes your command or, when there's more than one option available, it displays a list of suggested command actions/options/parameters.

When a suggestion list is displayed, depending on your shell, use the Tab or arrow keys to move through the list and press Enter to make a selection.

Examples

The following examples show various ways that you can use autocompletion:

Enter btp and press Tab to display all available actions:

àE

ã

./btp TAB

add

create

enable

list

logout

register

subscribe

unassign

unsubscribe

assign

delete

get

login

move

remove

target

unregister

update

àf

Partially enter btp cre and press Tab to autocomplete the command to btp create. Then, press Tab again to display a suggested list of group-object combinations:

àE

ã

```
./btp create TAB
accounts/directory
accounts/resource-provider
security/role
services/binding
accounts/environment-instance accounts/subaccount
security/role-collection
services/instance
àf
```

Partially enter a group and press Tab to display a suggested list of objects:

```
âE
ã
./btp create accountsTAB
accounts/directory
accounts/environment-instance accounts/resource-provider
accounts/subaccount
àf
```

Partially enter a parameter and press Tab to display a suggested list of parameters:

```
âE
ã
./btp create accounts/subaccount -TAB
--beta-enabled
--custom-properties
--description
--directory
--display-name
--global-account
--region
--subaccount-admins
--subdomain
--used-for-production
```

Procedureà^

Use btp enable autocomplete <SHELL> to enable command autocompletion for a specified shell.

The valid values for the supported shells are:

```
bash
powershell
zsh
```

Sample Code

```
btp enable autocomplete zsh
```

Enter the option of your choice.

To enable autocompletion, the startup file (RC file) of your shell is modified to load the available commands and parameters. The client looks for this RC file in your system and proposes possible files. You can choose to accept a proposal or specify a different RC file. To specify a custom path, first choose "Custom" and then enter the full path of the RC file to use. If no RC file is found, the clients prints an error message with the full path of the missing file so that you can create it and run the enable command again.

For example:

```
âE
ã
./btp enable autocomplete powershell
```

Which RCFile should be used for the installation?

1: C:\Documents\WindowsPowerShell\Microsoft.PowerShell\_profile.ps1

2: Custom

Enter option>2

Enter the full path of your RCFile>C:\Documents\WindowsPowerShell\Microsoft.PowerShell\_my-custom-

profile.ps1

àf

Start a new terminal session to activate the installed autocomplete script.

Resultsà^

The autocomplete option remains enabled in future sessions in your current client, until you disable it. To disable command autocomplete, run the following command:

âE

â

btp disable autocomplete <SHELL>

You can run either btp or btp --info to see if command autocomplete is currently enabled and where the autocomplete RC file for your shell is located. If you don't see a line specifying the location of the autocomplete script, then it's disabled.

Tip

If you see a discrepancy between the version of the autocomplete script and the client, the update of the autocomplete script might have failed. In such a case, try to disable and enable the autocomplete feature again. Whenever you start a new btp CLI terminal session, the autocomplete suggestions are automatically updated to include the latest commands.

If disabling the command autocomplete fails or you have uninstalled the btp CLI client without disabling autocomplete, you can manually remove traces of the autocomplete installation in your shell initialization file (RC or profile file depending on your shell) by deleting the section in your RC file from Begin SAP BTP command line interface autocomplete to End SAP BTP command line interface autocomplete.

---

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/enabling-events-exchange-between-sap-cloud-systems>

Title: Enabling Events Exchange Between SAP Cloud Systems | SAP Help Portal

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Enabling Events Exchange Between SAP Cloud Systems

Useà^

Note

The content in this section is not relevant for China (Shanghai) and Government Cloud (US) regions.

Systems that are included in Eventing Between SAP Cloud Systems formations on the System Landscape page of the SAP BTP cockpit can publish and consume events. The process of publishing and consuming events is in fact an exchange of event information across the customer system landscape and is driven by the system formations on the one hand side and the SAP Cloud Application Event Hub on the other.

Note

By creating a formation of type Eventing Between SAP Cloud Systems, keep in mind that there may be event distribution charges incurred if you exceed the free quota limit when using SAP Cloud Application Event Hub. To learn more about usage and consumption costs, see Usage and Consumption Costs.

Rulesà^

When creating Eventing Between SAP Cloud Systems formations, keep in mind the following rule:

Every system can be included in at most one Eventing Between SAP Cloud Systems formation.

If you register your SAP S/4HANA Cloud system using the All Communication Scenarios or Eventing Between SAP Cloud Systems communication scenario group when you get the registration token for this SAP S/4HANA Cloud system, the communication scenario SAP\_COM\_0892 will be enabled automatically after this SAP S/4HANA Cloud system is added to the formation of type Eventing Between SAP Cloud Systems in the Formations page. See Register an SAP S/4HANA Cloud System in a Global Account in SAP BTP.

Prerequisitesà^

You are a global account administrator, or you are a system landscape administrator. See Working with Role Collections.

You have subscribed to the SAP Cloud Application Event Hub application. See Subscribing to SAP Cloud Application Event Hub Application.

## Procedure

The following procedure outlines the steps you need to perform to enable the exchange of events across the systems within the system landscape.

In the SAP BTP cockpit, in the System Landscape Systems page of the SAP BTP cockpit, browse the already added systems in your customer system landscape or manually add and register any missing systems.

Every system you are going to include in this formation must not be part of another formation.

The customer system landscape features systems that are added to the list in one of the following ways:

### Auto-discovered

An auto-discovered system is a system (associated with the given global account) that has been discovered and added automatically to the list based on information of the existing system landscape. Any SAP system of the supported system types that is associated with the same customer ID, with which your global account in SAP BTP is associated, will be added automatically in the system landscape list.

### Subaccount/<my-subaccount>

A subscription in SAP BTP cockpit associated with a given subaccount. The subscription has been discovered and added automatically through the subaccount. Check the value of the Discovery column to see the subaccount where your system is subscribed.

### Manually added

Specifies that the system has been added to the list manually by the global account administrator, using the Add System button and completing the wizard. The system has been associated with the global account in SAP BTP.

### Note

If a given SAP system is missing on the Systems page, it may be associated with a different customer ID on the SAP BTP global account you are working in. In this case, you need to add the system manually, and then, register it.

### Tip

When you register an SAP S/4HANA Cloud system, you can automate some of the required configuration steps at a later point. To do this, you can use the All Communication Scenarios or Eventing Between SAP Cloud Systems communication scenario groups when you get the registration token for the SAP S/4HANA Cloud system. This allows the automatic enablement of the communication scenario SAP\_COM\_0892 after the corresponding system is added to the formation of type Eventing Between SAP Cloud Systems.

See Adding, Registering and Deregistering Systems.

Create a formation of type Eventing Between SAP Cloud Systems and include the relevant systems in it.

Add any name that helps you identify your formation.

In the Formation Type dropdown menu, select Eventing Between SAP Cloud Systems.

Select the systems that you want to include in the formation. One of these systems must be of type SAP Cloud Application Event Hub.

### Note

Systems can only be added to one formation of type Eventing Between SAP Cloud Systems in a global account.

Also, a formation of type Eventing Between SAP Cloud Systems can contain only one system of type SAP Cloud Application Event Hub.

Review your selections and create the formation.

## Next Steps

When the formation is created, navigate to the SAP Cloud Application Event Hub application and enable the event subscriptions. See Enabling SAP Event Subscriptions.

## Remove a System from the Formation

If you want to remove an SAP system from a formation of type Eventing Between SAP Cloud Systems, make sure that all active event subscriptions are disabled first in SAP Cloud Application Event Hub. You can check your subscriptions in the SAP Cloud Application Event Hub application on the Subscriptions page. See Disabling SAP Event Subscriptions.

---

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/environment-variables>

Title: Environment Variables | SAP Help Portal

HomeSAP Business Technology Platform (SAP BTP)SAP Business Technology Platform...Application RouterEnvironment Variables

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## Environment Variables

A list of environment variables that can be used to configure the application router.

The following table lists the environment variables that you can use to configure the application router. The table also provides a short description of each variable and, where appropriate, an example of the configuration data.

### Application Router Configuration Variables

| Variable | Description |
|----------|-------------|
|----------|-------------|

|             |                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------|
| httpHeaders | Configures the application router to return additional HTTP headers in its responses to client requests |
|-------------|---------------------------------------------------------------------------------------------------------|

|              |                                                                                   |
|--------------|-----------------------------------------------------------------------------------|
| destinations | Provides information about the available application (microservice) destinations. |
|--------------|-----------------------------------------------------------------------------------|

|         |                                                                                                  |
|---------|--------------------------------------------------------------------------------------------------|
| cookies | The application router generates the following third-party cookies and sends them to the client: |
|---------|--------------------------------------------------------------------------------------------------|

|                    |  |
|--------------------|--|
| locationafterLogin |  |
|--------------------|--|

|                    |  |
|--------------------|--|
| fragmentAfterLogin |  |
|--------------------|--|

|           |  |
|-----------|--|
| signature |  |
|-----------|--|

|             |                                                                                                                          |
|-------------|--------------------------------------------------------------------------------------------------------------------------|
| JSESSION ID | To control the behavior and usage of the third-party cookies, you can configure the attributes SameSite and Partitioned. |
|-------------|--------------------------------------------------------------------------------------------------------------------------|

For more information about third-party cookies, please see the KBA 3409306 .

|                 |                                                                                                              |
|-----------------|--------------------------------------------------------------------------------------------------------------|
| SESSION_TIMEOUT | Sets the time to trigger an automatic central log out from the User Account and Authentication (UAA) server. |
|-----------------|--------------------------------------------------------------------------------------------------------------|

|                    |                                            |
|--------------------|--------------------------------------------|
| SEND_XFRAMEOPTIONS | Sets or changes the X-Frame-Options header |
|--------------------|--------------------------------------------|

|                      |                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------|
| CJ_PROTECT_WHITELIST | A list of allowed server or domain origins to use when checking for click-jacking attacks. |
|----------------------|--------------------------------------------------------------------------------------------|

|                    |                                                                                                       |
|--------------------|-------------------------------------------------------------------------------------------------------|
| WS_ALLOWED_ORIGINS | A list of the allowed server (or domain) origins that the application router uses to verify requests. |
|--------------------|-------------------------------------------------------------------------------------------------------|

|             |                                                                                                                                                                      |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| JWT_REFRESH | Configures the automatic refresh of the JSON Web Token (JWT) provided by the User Account and Authentication (UAA) service to prevent expiry (default is 5 minutes). |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                  |                                                                        |
|------------------|------------------------------------------------------------------------|
| UAA_SERVICE_NAME | Specifies the exact name of the UAA service to bind to an application. |
|------------------|------------------------------------------------------------------------|

|                             |                                                                                                                                    |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| INCOMING_CONNECTION_TIMEOUT | Specifies the maximum time (in milliseconds) for a client connection. If the specified time is exceeded, the connection is closed. |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------|

|                     |                                                                                                  |
|---------------------|--------------------------------------------------------------------------------------------------|
| TENANT_HOST_PATTERN | Defines a regular expression to use when resolving tenant host names in the request's host name. |
|---------------------|--------------------------------------------------------------------------------------------------|

|             |                                                                          |
|-------------|--------------------------------------------------------------------------|
| COMPRESSION | Configures the compression of resources before a response to the client. |
|-------------|--------------------------------------------------------------------------|

|                       |                                                                                           |
|-----------------------|-------------------------------------------------------------------------------------------|
| SECURE_SESSION_COOKIE | Configures the enforcement of the Secure flag of the application router's session cookie. |
|-----------------------|-------------------------------------------------------------------------------------------|

|               |                                                                  |
|---------------|------------------------------------------------------------------|
| REQUEST_TRACE | Enables additional traces of the incoming and outgoing requests. |
|---------------|------------------------------------------------------------------|

|                        |                                                                                                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EXTERNAL_REVERSE_PROXY | Indicates the use of the application router behind an external reverse proxy outside of the Cloud Foundry domain. For more information, see Customer Header under Headers. |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|      |                                                                                                              |
|------|--------------------------------------------------------------------------------------------------------------|
| CORS | Provides support for cross-origin requests, for example, by allowing the modification of the request header. |
|------|--------------------------------------------------------------------------------------------------------------|

|                             |  |
|-----------------------------|--|
| DIRECT_ROUTING_URI_PATTERNS |  |
|-----------------------------|--|

Defines a list of URIs that are directed to the routing configuration file (xs-app.json file) of the application router instead of to a specific application's xs-app.json file that is stored in the HTML5 Application Repository. This configuration improves the application loading time and monitoring options.

#### EXT\_SESSION\_MGT

You can configure external session management. See External Session Management.

#### CF\_NODEJS\_LOGGING\_LEVEL

Sets the minimal logging level of the cf-nodejs-logging-support library of the application router.

#### STATE\_PARAMETER\_SECRET

Enables the use of state parameters to prevent CRFS attacks.

If this environment variable is set, the application router creates a state parameter for each initial authorization request. By validating that the authentication server returns the same state parameter in its response, the application server can verify that the response did not originate from a third party.

#### HTTP2\_SUPPORT

Enables the application router to start as an HTTP/2 server.

#### Note

To configure HTTP/2 support, you must use Cloud Foundry routes with an HTTP/2 destination protocol. See Configuring HTTP/2 Support in the Cloud Foundry Documentation.

As connection-specific header fields aren't supported by the HTTP/2 protocol (see <https://datatracker.ietf.org/doc/html/rfc9113>), the application router removes such headers automatically when they are returned from a backend to prevent a failure of the HTTP/2 response.

#### FULL\_CERTIFICATE\_CHAIN

Enables the application router to send the entire chain of certificates provided in the binding of the authentication service (XSUAA or Identity Authentication) to the backend applications. The certificates are required for mutual TLS authentication (mTLS handshake). See Mutual TLS Authentication (mTLS) and Certificates Handling.

#### OWN\_SAP\_CLOUD\_SERVICE

An array that contains the business solutions ("SAP cloud services") that your HTML5 applications are associated with as values. This configuration enables the standalone application router to use the same standardized format for runtime URLs (`/<sap.cloud.service>.<appId>-<versionId>/`) that is also used by the managed application router. If a runtime URL contains one of the defined values in the `<sap.cloud.service>` section, the application router will recognize the value during the processing of a request.

#### DYNAMIC\_IDENTITY\_PROVIDER

Configures a dynamic identity provider.

If `DYNAMIC_IDENTITY_PROVIDER` is set to `true`, the end user can set the identity provider (IDP) for the application's login process. If SAP Authorization and Trust Management is used, the `sap_idprequest` query parameter must be filled with the IDP origin key. If Identity Authentication is used, the `sap_idp` request query parameter must be filled with the name of the corporate identity provider configured in the administration console for SAP Cloud Identity services. Note that this is not the display name, but the name of the corporate identity provider (see Creating URL To Access Application with Specific Identity Provider ).

If the `identityProvider` property is defined in the route, its value will be overwritten by the `sap_idp` query parameter value.

The default value for `DYNAMIC_IDENTITY_PROVIDER` is `false`.

#### httpHeaders<sup>a</sup>

If configured, the application router sends additional HTTP headers in its responses to a client request. You can set the additional HTTP headers in the `<httpHeaders>` environment variable. The following example configuration shows how to configure the application router to send two additional headers in the responses to the client requests from the application `<myApp>`:

```
cf set-env <myApp> httpHeaders "[ { \"X-Frame-Options\": \"ALLOW-FROM http://example.com\" }, { \"Test-Additional-Header\": \"1\" } ]"
```

or

```
cf set-env <myApp> httpHeaders '[ { "X-Frame-Options": "ALLOW-FROM http://example.com" }, { "Test-Additional-Header": "1" } ]'
```

#### Tip

To ensure better security of your application set the Content-Security-Policy header. This is a response header

which informs browsers (capable of interpreting it) about the trusted sources from which an application expects to load resources. This mechanism allows the client to detect and block malicious scripts injected into the application. A value can be set via the <httpHeaders> environment variable in the additional headers configuration. The value represents a security policy which contains directive-value pairs. The value of a directive is an allowlist of trusted sources.

Usage of the Content-Security-Policy header is considered second line of defense. An application should always provide proper input validation and output encoding.

destinationsâ

The destinations configuration is an array of objects that is defined in the destinations environment variable. A destination is required for every application (microservice) that is part of the business application. The following table lists the properties that can be used to describe the destination:

#### Destination Environment Variable Properties

| Property         | Type    | Mandatory | Description                                                                                                                                                                                                                                                                                                                                      |
|------------------|---------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| name             | String  | Yes       | A unique identifier for the destination                                                                                                                                                                                                                                                                                                          |
| url              | String  | Yes       | The Unique Resource Locator for the application (microservice)                                                                                                                                                                                                                                                                                   |
| proxyHost        | String  | No        | The host of the proxy server used in case the request should go through a proxy to reach the destination.                                                                                                                                                                                                                                        |
| proxyPort        | String  | No        | The port of the proxy server used in case the request should go through a proxy to reach the destination.                                                                                                                                                                                                                                        |
| forwardAuthToken | Boolean | No        | If true, the OAuth token will be sent to the destination. The default value is "false". This token contains the user identity, scopes, and some other attributes. The token is signed by the User Account and Authorization (UAA) service so that the token can be used for user-authentication and authorization purposed by back-end services. |
| strictSSL        | Boolean | No        | Configures whether the application router should reject untrusted certificates. The default value is "true".                                                                                                                                                                                                                                     |
| timeout          | Number  | No        | A positive integer representing the maximum amount of time to wait for a response (in milliseconds) from the specified destination. The default is 30000ms.                                                                                                                                                                                      |

The timeout specified here also applies to the logout path, `logoutPath`, if the logout path is defined, for example, in the application's descriptor file `xs-app.json`.

`proxyType`

String

No

Indicates if the destination is used to access applications in private networks (for example, on-premise) or publicly available on the Internet. Possible value: `onPremise`. If `proxyType` is not specified, the default (Internet access) is assumed.

Tip

In Cloud environments, if you set the application's destination `proxyType` to `onPremise`, a binding to the SAP BTP connectivity service is required, and the `forwardAuthToken` property must not be set.

`IASDependencyName`

String

No

Configures the name of the Identity Authentication dependency that is used to exchange the Identity Authentication token used for the user authentication at login. If configured, the exchanged token is then also forwarded to the backend applications.

`forwardAuthCertificates`

Boolean

No

If true, the certificates and key of the authentication service are added to the HTTP connection to the destination. The default value is false. For more information see: Mutual TLS Authentication (mTLS) and Certificates Handling

If true, the certificates and key of the authentication service are added to the HTTP connection to the destination. The default value is false. For more information see: Mutual TLS Authentication (mTLS) and Certificates Handling

The following example shows a simple configuration for the `<destinations>` environment variable:

Sample Code

⌵

⌵

```
[
  {
    "name" : "ui5",
    "url" : "https://sapui5.example.com",
    "proxyHost" : "proxy",
    "proxyPort" : "8080",
    "forwardAuthToken" : false,
    "timeout" : 1200
  }
]
```

It is also possible to include the destinations in the `manifest.yml` file, as illustrated in the following example:

Sample Code

⌵

⌵

```
- name: node-hello-world
memory: 100M
path: web
env: destinations: >
[
  {"name":"ui5", "url":"https://sapui5.example.com"}
]
```

Additional Cookie Attributes⌵

If configured, the application router sends additional cookie attributes in its responses to the client. Additional cookies attributes can be set in the `<COOKIES>` environment variable.

Example of configuration for cookies in the manifest.yml:

Sample Code

```
âE
â
env:
COOKIES: >
{
"SameSite": "None",
"Partitioned":
{
"supportedPartitionAgents": "^Mozilla.*(Chrome|Chromium)/((109)|(1[1-9][0-9])|([2-9][0-9][0-9]))",
"unsupportedPartitionAgents": "PostmanRuntime/7.29.2"
}
}
}
âf
```

In this example, the application router sets the SameSite attribute of the cookie to None and the specifies a Partitioned attribute that is sent in the responses to the client.

Note

Currently, only the values None and Lax are supported for the SameSite attribute. The value Strict is not supported.

The Partitioned attribute contains two required regular expressions:

supportedPartitionAgents for supported agents

unsupportedPartitionAgents for unsupported agents

You can use a wildcard '.\*' in supportedPartitionAgents to allow all agents to use the Partitioned attribute.

Note

unsupportedPartionAgents overwrites the configurations in supportedPartitionAgents. If an agent is allowed in supportedPartitionAgents but disallowed in unsupportedPartitionAgents, the Partitioned attribute will not be returned.

SESSION\_TIMEOUTâ^

You can configure the triggering of an automatic central log-out from the User Account and Authentication (UAA) service if an application session is inactive for a specified time. A session is considered to be inactive if no requests are sent to the application router. The following command shows how to set the environment variable <SESSION\_TIMEOUT> to 40 (forty) minutes for the application <myApp1>:

```
cf set-env <myApp1> SESSION_TIMEOUT 40
```

Note

You can also set the session timeout value in the application's manifest.yml file, as illustrated in the following example:

Sample Code

```
âE
â
- name: myApp1
memory: 100M
path: web
env:
SESSION_TIMEOUT: 40
```

Tip

If the authentication type for a route is set to "xsuaa" (for example, "authenticationType": "xsuaa"), the application router depends on the UAA server for user authentication, and the UAA server might have its own session timeout defined. To avoid problems caused by unexpected timeouts, it is recommended that the session timeout values configured for the application router and the UAA are identical."

SEND\_XFRAMEOPTIONSâ^

By default, the application router sends the X-Frame-Options header with the value "SAMEORIGIN". You can change this behavior either by disabling the sending of the default header value (for example, by setting SEND\_XFRAMEOPTIONS environment variable to false) or by overriding the value, for example, by configuring

additional headers (with the <httpHeaders> environment variable).

The following example shows how to disable the sending of the X-Frame-Options for a specific application, myApp1:

```
cf set-env <myApp1> SEND_XFRAMEOPTIONS false
CJ_PROTECT_WHITELIST^
```

The <CJ\_PROTECT\_WHITELIST> specifies a list of origins (for example, host or domain names) that do not need to be protected against click-jacking attacks. This list of allowed host names and domains are used by the application router's allowlist service to protect XS advanced applications against click-jacking attacks. When an HTML page needs to be rendered in a frame, a check is done by calling the allowlist service to validate if the parent frame is allowed to render the requested content in a frame. The check itself is provided by the allowlist service

The following example shows how to add a host name to the click-jacking protection allowlist for the application, myApp1:

```
cf set-env <myApp1> CJ_PROTECT_WHITELIST {<protocol>, <hostname>, <portNr>}
```

The content is a JSON list of objects with the properties listed in the following table:

Allowlist of Host and Domain Names

Property

Type

Mandatory

Description

protocol

String

No

URI scheme, for example "HTTP".

host

String

Yes

A valid host name, for example, example.com.hostname, or a domain name defined with an asterisk (\*)

\*.example.com.

port

String/Number

No

Port string or number containing a valid port.

Note

Matching is done against the properties provided. For example, if only the host name is provided, the allowlist service matches all schemata and protocols.

```
xs set-env <myApp1> CJ_PROTECT_WHITELIST {<*.example.com>}
```

```
WS_ALLOWED_ORIGINS^
```

When the application router receives an upgrade request, it verifies that the origin header includes the URL of the application router. If this is not the case, then an HTTP response with status 403 is returned to the client.

This origin verification can be further configured with the environment variable <WS\_ALLOWED\_ORIGINS>, which defines a list of the allowed origins the application router uses in the verification process.

Note

The structure of the <WS\_ALLOWED\_ORIGINS> variable is the same as the variable

<CJ\_PROTECT\_WHITELIST>.

```
cf set-env <myApp1> WS_ALLOWED_ORIGINS {<*.example.com>}
```

```
JWT_REFRESH^
```

The JWT\_REFRESH environment variable is used to configure the application router to refresh a JSON Web Token (JWT) for an application, by default, 5 minutes before the JWT expires, if the session is active.

```
cf set-env <myApp1> JWT_REFRESH 1
```

If the JWT is close to expiration and the session is still active, a JWT refresh will be triggered

<JWT\_REFRESH> minutes before expiration. The default value is 5 minutes. To disable the automatic refresh, set the value of <JWT\_REFRESH> to 0 (zero).

```
UAA_SERVICE_NAME^
```

The UAA\_SERVICE\_NAME environment variable enables you to configure an instance of the User Account and Authorization service for a specific application, as illustrated in the following example:

```
cf set-env <myApp1> UAA_SERVICE_NAME <myUAAServiceName>
```

#### Note

The details of the service configuration are defined in the <VCAP\_SERVICES> environment variable, which is not configured by the user.

#### INCOMING\_CONNECTION\_TIMEOUT

The INCOMING\_CONNECTION\_TIMEOUT environment variable enables you to set the maximum time (in milliseconds) allowed for a client connection, as illustrated in the following example:

```
cf set-env <myApp1> INCOMING_CONNECTION_TIMEOUT 60000
```

If the specified time is exceeded, the connection is closed. If INCOMING\_CONNECTION\_TIMEOUT is set to zero (0), the connection-timeout feature is disabled. The default value for INCOMING\_CONNECTION\_TIMEOUT is 120000 ms (2 min).

#### TENANT\_HOST\_PATTERN

The TENANT\_HOST\_PATTERN environment variable enables you to specify a string containing a regular expression with a capturing group. The requested host is matched against this regular expression. The value of the first capturing group is used as the tenant Id. as illustrated in the following example:

```
cf set-env <myApp1> TENANT_HOST_PATTERN
```

#### COMPRESSION

The COMPRESSION environment variable enables you to configure the compression of resources before a response to the client, as illustrated in the following example:

```
cf set-env <myApp1> COMPRESSION
```

Here is a complete example of the compression environment variable:

#### Sample Code

```
âE
```

```
ã
```

```
env:
```

```
COMPRESSION: >
```

```
{
  "enabled": true,
  "minSize": 2048,
  "compressResponseMixedTypeContent": true
}
```

#### SECURE\_SESSION\_COOKIE

The SECURE\_SESSION\_COOKIE can be set to true or false. By default, the Secure flag of the session cookie is set depending on the environment the application router runs in. For example, when the application router is running behind a router that is configured to serve HTTPS traffic, then this flag will be present. During local development the flag is not set.

#### Note

If the Secure flag is enforced, the application router will reject requests sent over unencrypted connections.

The following example illustrates how to set the SECURE\_SESSION\_COOKIE environment variable:

```
cf set-env <myApp1> SECURE_SESSION_COOKIE true
```

#### REQUEST\_TRACE

You can enable additional traces of the incoming and outgoing requests by setting the environment variable <REQUEST\_TRACE>“true”. If enabled, basic information is logged for every incoming and outgoing request to the application router.

The following example illustrates how to set the REQUEST\_TRACE environment variable:

```
cf set-env <myApp1> REQUEST_TRACE true
```

#### Tip

This is in addition to the information generated by the Node.js package @sap/logging that is used by the XS advanced application router.

#### CORS

The CORS environment variable enables you to provide support for cross-origin requests, for example, by allowing the modification of the request header. Cross-origin resource sharing (CORS) permits Web pages

from other domains to make HTTP requests to your application domain, where normally such requests would automatically be refused by the Web browser's security policy.

Cross-origin resource sharing (CORS) is a mechanism that allows restricted resources on a Web page to be requested from another domain (protocol and port) outside the domain (protocol and port) from which the first resource was served. The CORS configuration enables you to define details to control access to your application resource from other Web browsers. For example, you can specify where requests can originate from or what is allowed in the request and response headers. The following example illustrates a basic CORS configuration:

```
â€”
{
  "uriPattern": "\\route1$",
  "allowedMethods": [
    "GET"
  ],
  "allowedOrigin": [
    {
      "host": "host.example.com",
      "protocol": "https",
      "port": 345
    }
  ],
  "maxAge": 3600,
  "allowedHeaders": [
    "Authorization",
    "Content-Type"
  ],
  "exposeHeaders": [
    "customHeader1",
    "customHeader2"
  ],
  "allowedCredentials": true
}
```

The CORS configuration includes an array of objects with the following properties, some of which are mandatory:

Available Settings for CORS Options

CORS Property

Type

Mandatory

Description

uriPattern

String

Yes

A regular expression (RegExp) representing the source routes to which the CORS configuration applies. To ensure that the RegExp matches the complete path, surround it with ^ and \$,f or example, "uriPattern":

"^\\route1\$". Defaults: none

allowedOrigin

Array

Yes

A comma-separated list of objects each of which contains a host name, port and protocol that are allowed by the server, for example: [{"host": "www.example.com"}] or [{"host": ".example.com"}].

Note

Matching is case-sensitive. In addition, if no port or protocol is specified, the default is "\*".

The default configuration is: [{"host": "\*"}], which means that the server allows any origin to access the resource.

allowedMethods

Array

No

A comma-separated list of HTTP methods that are allowed by the server, for example, "GET", "POST". If allowMethods is defined but no method is specified, the default "GET", "POST", "HEAD", "OPTIONS" (all) applies.

Tip

The specified methods must be upper-case, for example, GET. Matching of the method type is case-sensitive.

allowedHeaders

Array

No

A comma-separated list of request headers that are allowed by the server. the default values are as follows: ["Origin", "Accept", "X-Requested-With", "Content-Type", "Access-Control-Request-Method", "Access-Control-Request-Headers"].

maxAge

String

No

A single value specifying the length of time (in seconds) a preflight request should be cached for. A negative value that prevents CORS filter from adding this response header to the pre-flight response. If maxAge is defined but no value is specified, the default time of "1800" seconds applies.

exposeHeaders

Array

No

A comma-separated list of response headers that are allowed to be exposed. If exposeHeaders is defined but no response header is specified for exposure, no default value is supplied.

allowedCredentials

Boolean

No

A Boolean flag that indicates whether the specified resource supports user credentials. The default setting is "true".

It is also possible to include the CORS configuration in either the manifest.yml or the manifest-op.yml file. The code in the following example enables the CORS configuration for any route with a source URI pattern that matches the RegExp "^/route1\$":

Sample Code

CORS Configuration in the manifest.yml File

âE

ã

- name: node-hello-world

memory: 100M

path: web

env:

CORS: >

```
[
{
  "allowedOrigin": [
    {
      "host": "my_host",
      "protocol": "https"
    }
  ],
  "uriPattern": "^/route1$"
}
```

## DIRECT\_ROUTING\_URI\_PATTERNS

With the direct routing URI patterns configuration, you can define a list of URIs that are directed to the routing configuration file (xs-app.json file) of the application router instead of to a specific application's xs-app.json file that is stored in the HTML5 Application Repository. This configuration improves the application loading time and monitoring options because it prevents unnecessary calls to the HTML5 Application Repository.

The configuration is an array of strings or regular expressions. You have to provide only the first segment in the URL, after the approuter host. For example, for the URL `https://<approuter-host>/route1/index.html`, you enter `route1` in the direct routing URI patterns array.

Sample Code

⌕

⌕

env:

```
DIRECT_ROUTING_URI_PATTERNS: >
```

```
["route1", "^route2$", "route3"]
```

## CF\_NODEJS\_LOGGING\_LEVEL

With the `CF_NODEJS_LOGGING_LEVEL` variable, you can set the minimal logging level of the `cf-nodejs-logging-support` library of the application router. The following levels are available:

off

error

warn

info

verbose

debug

silly

The default value is `error`.

Here is a sample content for the `CF_NODEJS_LOGGING_LEVEL` environment variable:

Sample Code

⌕

⌕

env:

```
CF_NODEJS_LOGGING_LEVEL: "debug"
```

## Related Information

Application Router

Routing Configuration File

---

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/fc2bf6a63c254b92afb3356e2df0f930.html>

Title: *Getting Support, Neo Environment | SAP Help Portal*

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Getting Support, Neo Environment

To get assistance, use the available support channels provided by SAP for Me.

## Caution

SAP Business Technology Platform, Neo environment will sunset on December 31, 2028, subject to terms of customer or partner contracts.

For more information, see SAP Note 3351844 .

## Tip

This documentation refers to SAP Business Technology Platform, Neo environment. If you are looking for documentation about other environments, see SAP Business Technology Platform .

## Prerequisites

Before you report a case, ensure the following:

Check Platform Status

Visit SAP Trust Center. You can check:

availability by service on the Cloud Status tab page;

availability by region on the Data Center tab page.

#### Check Tools Versions

Make sure that you've the latest versions of the tools you're using (recommended), or at least the versions you're using are still supported. For more information, see SAP Development Tools.

#### Try Out Built-In Support Within SAP BTP Cockpit

When using SAP BTP cockpit as an S-user, try Built-In Support to get customer help within the cockpit. See Built-In Support.

#### Learn the Best Practices and Offerings

Familiarize with the guidelines how to engage with support. See Support from SAP Best Practices and Offerings.

#### Procedureà^

To report a case in SAP for Me, proceed as follows:

Open SAP for Me and log on.

Open the Services & Support dashboard.

(Optional) You can use the Knowledge Search tab for existing solutions in:

SAP Notes & KBAs

SAP Community

SAP Support Portal

Product Documentation

Use the Get Support application within SAP for Me to get assistance via the available support channels.

For more information about the specifics of each support channel and to learn how to report problems, see Get Support Application.

#### Creating a Support User

Sometimes you might need to create a support user to troubleshoot issues in your system. For more information about how and when to create a support user, see 3543065 - Best Practices for Creating Support Users in Your BTP System.

#### Related Information

Users and Authorizations for SAP One Support Launchpad

Getting Support for the Multi-Cloud Environment

Cloud Availability Center

Road Map for Neo Environment

---

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/manage-application-instances>

Title: Manage Application Instances | SAP Help Portal

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#### Manage Application Instances

To increase the availability of your Cloud Foundry application, you can run multiple instances of it. In the SAP BTP cockpit, you can add or remove application instances, or change the size of their memory and disk.

#### Prerequisitesà^

You must have one application deployed in your Cloud Foundry space.

You have the Space Developer or Space Supporter role.

#### Procedureà^

In the SAP BTP cockpit, navigate to the Applications page in your Cloud Foundry space.

Choose the application whose instances you want to manage.

On the Application Overview page, you can:

Change Configuration to update the memory size and the disk size of all instances of your application.

#### Note

Each instance of the application receives the size in megabytes you have specified. You can allocate instance memory up to the available space memory limit.

Bind Application Autoscaler to scale your application by binding it to the Application Autoscaler service. Once

it's bound, you can access the Application Autoscaler dashboard from the Instances Configuration section.

Add Instance to start an additional instance of your application.

Remove Instance to remove one of the running instances of your application.

Related Information

Deploy an Application

---

URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/navigate-to-orgs-and-spaces>

Title: SAP Help Portal | SAP Online Help

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/org-management-using-sap-btp-command-line-interface-btp-cli>

Title: Org Management Using the SAP BTP Command Line Interface (btp CLI) | SAP Help Portal

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OperationsAdministration and Operations in the Cloud Foundry EnvironmentOrg Management Using the SAP

BTP Command Line Interface (btp CLI)

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Org Management Using the SAP BTP Command Line Interface (btp CLI)

The Cloud Foundry environment allows you to create polyglot cloud applications in Cloud Foundry. To manage

the lifecycle of an org in the Cloud Foundry environment, use the `accounts/environment-instance` command in the `bt` CLI.

#### Available Plans

To create a Cloud Foundry org in a subaccount, you use the Cloud Foundry Runtime service. The Cloud Foundry Runtime service offers the following plans for this purpose:

##### Plan Name

##### Description

##### standard

This is an enterprise-grade plan that allows you to create an org in your Cloud Foundry environment to start developing polyglot cloud-native applications.

By default, a Cloud Foundry org that is created with this plan does not have any application runtime. To assign application runtime, the global account admin must assign the Cloud Foundry Runtime service with the MEMORY plan to the subaccount so that the org has memory for its applications.

##### free

Use this plan to try out the Cloud Foundry environment without an additional charge before switching to the standard plan that supports enterprise-grade productivity.

By default, a Cloud Foundry org that is created with this plan is given a fixed application-runtime quota. There is no need to assign the Cloud Foundry Runtime service with the MEMORY plan to the subaccount as it will have no effect on the org quota.

For more information about the free tier model for SAP BTP and its availability, see [Using Free Service Plans](#).

##### trial

Use this plan to utilize your trial period.

##### Note

Your subaccount's type must be trial to use this plan.

For more information, about the features that each plan offers, see [Cloud Foundry Runtime on SAP Discovery Center](#).

#### Command Reference for CRUD Operations

The `bt` CLI offers the following environment instance command actions to manage Cloud Foundry orgs in SAP BTP:

##### Action

##### Command

##### Description

##### Parameters

##### Additional Info

##### list

```
bt list accounts/environment-instance --subaccount <SUBACCOUNT_ID>
```

List all the Cloud Foundry orgs and other environment instances in a subaccount.

SUBACCOUNT\_ID: The ID of the subaccount.

##### get

```
bt get accounts/environment-instance <ID> --subaccount <SUBACCOUNT_ID>
```

Get the details of a specific Cloud Foundry org in a subaccount.

ID: The ID of the environment instance to view.

SUBACCOUNT\_ID: The ID of the subaccount.

##### create

```
bt create accounts/environment-instance --subaccount <SUBACCOUNT_ID> --display-name  
<DISPLAY_NAME> --service <SERVICE> --plan <PLAN> --environment cloudfoundry --parameters  
{"instance_name":"<ORG_NAME>"}
```

##### Note

If the logged-in region has multiple landscapes, then you must specify the name of the landscape using the `--landscape` parameter in the command. You can obtain the valid values using the `bt list accounts/available-environment` command.

Create a Cloud Foundry org in a subaccount.

SUBACCOUNT\_ID: The ID of the subaccount in which to create the Cloud Foundry org.

DISPLAY\_NAME: The SAP BTP name of the environment instance.

PLAN: Specify either standard or free as the value.

ORG\_NAME: The name of the org in the Cloud Foundry environment. Spaces are not allowed in the org name. Once the org is created, you cannot change its name.

For examples that show how to pass JSON parameters in the command line with different operating systems and shells, see the Passing JSON Parameters on the Command Line.

update

`btpt update accounts/environment-instance <ID> --subaccount <SUBACCOUNT_ID> --plan <PLAN>`

Update an existing Cloud Foundry org in a subaccount.

ID: The ID of the environment instance to update

SUBACCOUNT\_ID: The ID of the subaccount.

PLAN: Specify either standard or free as the value.

For the Cloud Foundry environment, you can use this command to change only the plan of an existing Cloud Foundry environment instance. In other words, update an org created with the free plan of the Cloud Foundry Runtime to the standard plan (not vice versa).

Note

Before updating the plan to standard, make sure that the MEMORY plan for the Cloud Foundry Runtime entitlement is assigned to your subaccount, and that you've assigned sufficient application runtime quota to cover the current workload in the subaccount.

delete

`btpt delete accounts/ environment-instance <ID> --subaccount <SUBACCOUNT_ID>`

Delete a Cloud Foundry org created in a subaccount.

ID: The ID of the environment instance to delete.

SUBACCOUNT\_ID: The ID of the subaccount where the environment instance exists.

The Cloud Foundry org and all its data will be lost.

Note

You can obtain the subaccount ID by running `btpt list accounts/subaccount`. You can also target the subaccount using `btpt target --subaccount <ID>`, and then omit the subaccount parameter from the rest of the commands.

Adding and Removing Org Managers

You can also use the `btpt` CLI to add or remove org managers.

Action

Command

Description

Parameters

Additional Info

update

For the default identity provider:

`btpt update accounts/environment-instance <ID> --subaccount <SUBACCOUNT_ID> --plan <PLAN> --parameters '{"usersToAdd":[{"id":"myUserID"},"email":"name@example.com"]}]'`

For a custom identity provider:

`btpt update accounts/environment-instance <ID> --subaccount <SUBACCOUNT_ID> --plan <PLAN> --parameters '{"usersToAdd":[{"id":"myUserID"},"email":"name@example.com","origin":"<your custom IDP>"}]}'`

Add one or more org managers to a Cloud Foundry org in a subaccount.

`usersToAdd` adds the OrgManager role to the user, and, if it doesn't exist yet, it creates the user in the org.

ID: The ID of the environment instance for which to add org managers.

SUBACCOUNT\_ID: The ID of the subaccount.

PLAN: Specify either standard or free as the value.

For examples that show how to pass JSON parameters in the command line with different operating systems and shells, see the Passing JSON Parameters on the Command Line.

update

For the default identity provider:

`btpt update accounts/environment-instance <ID> --subaccount <SUBACCOUNT_ID> --plan <PLAN> --parameters '{"usersToRemove":[{"id":"myUserID"},"email":"name@example.com"]}]'`

For a custom identity provider:

```
btp update accounts/environment-instance <ID> --subaccount <SUBACCOUNT_ID> --plan <PLAN> --
parameters "{\"usersToRemove\":[{\"id\":\"myUserID\",\"email\":\"name@example.com\",\"origin\":\"<your custom
IDP>\"}]}"
```

Remove one or more org managers from a Cloud Foundry org in a subaccount.

usersToRemove removes the OrgManager role from the user, but it doesn't remove the user from the org.

ID: The ID of the environment instance for which to remove org managers.

SUBACCOUNT\_ID: The ID of the subaccount.

PLAN: Specify either standard or free as the value.

To learn how to remove a user from the Cloud Foundry org and the subaccount, see [About User Management in the Cloud Foundry Environment](#).

Related Information

[Account Administration Using the SAP BTP Command Line Interface \(btp CLI\)](#)

[About Roles in the Cloud Foundry Environment](#)

[Org Administration Using the Cockpit](#)

[Add Org Members](#)

[Managing Spaces](#)

[About User Management in the Cloud Foundry Environment](#)

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/platform-updates-and-notifications>

Title: [Platform Updates and Notifications | SAP Help Portal](#)

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[Platform Updates and Notifications](#)

SAP BTP is a dynamic product, which has continuous production releases (updates). To get notifications for the new features and fixes every release, subscribe at [What's New for SAP Business Technology Platform](#). For more information, see [Subscribing to What's New Notifications](#).

Reasons for Region Updates

Regions are updated in the following cases:

Biweekly updates (standard) - aligned with the contractual obligations to customers and partners. Such updates usually don't affect productive applications, because most services support zero downtime maintenance. For more information about the biweekly updates, see [Consolidated Release Schedules for SAP BTP, Intelligent Enterprise Suite: Harmonized release calendar for SAP Cloud products](#), and [Service Level Agreement for SAP Cloud Services](#).

Immediate updates - fixes required for bugs that affect productive application operations, or due to urgent security fixes. In some cases, this might lead to downtime or application restart, for which the application groups receive a notification.

Major upgrades - happen rarely, in a bigger maintenance window, up to four times per year. For the time frames of the services' major upgrades, see [Service Level Agreement for SAP Cloud Services](#). We let you know about these upgrades 4 weeks in advance.

Announcements and Subscriptions [AWS, Azure, or GCP Regions]

You can follow the availability of the platform at [SAP Trust Center](#). You can check:

the availability by service on the [Cloud Status](#) tab page;

the availability by region on the [Data Center](#) tab page.

In addition, you can use [Cloud Availability Center \(CAC\)](#) to check what cloud products and services are assigned to your customer ID for notifications. CAC provides you with a personalized dashboard based on SAP Fiori. It's designed to give you quick access to relevant information about the cloud products that you own and their availability and maintenance status. For more information, see [CAC User Guide](#).

You can find information about the supported Cloud Product Cloud Service at [Supported Products](#).

Note

The foundational components of the platform are included in [SAP Extension Suite – Development Efficiency Foundational Services for SAP BTP](#).

To get notifications for updates and downtimes, create a subscription by specifying Cloud Product, Cloud

Service, and Notification Type. For more information, see Cloud System Notification Subscriptions User Guide.

#### Example

To receive notifications about changes in the IP ranges of a region, subscribe to SAP Extension Suite – Development Efficiency Foundational Services for SAP BTP.

#### Example

To receive notifications about what application processes to restart to apply new security patches, subscribe to SAP Extension Suite – Development Efficiency Foundational Services for SAP BTP.

Announcements and Subscriptions [China (Shanghai) Region]

You can follow the availability of the platform and the announcements about upcoming updates and downtimes at China (Shanghai) region status page. Subscribe to the status page to get notifications for updates and downtimes.

Announcements and Subscriptions [Government Cloud (US) Region]

Planned downtimes and outage communication are sent through e-mail to the initial administrator of your global account.

Related Information

What's New

Regions

Release Schedule for cross and foundational services of SAP BTP, Cloud Foundry environment

Release Schedule for SAP BTP, Kyma Runtime

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/process-findings-after-analysis>

Title: Process Findings After Analysis | SAP Help Portal

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Process Findings After Analysis

Find out how you can process your findings after custom code analysis.

Purpose

In case you've been wondering what the next steps would be after custom code analysis and what to do with your findings in the Analyze Custom Code app, this topic is for you. Here, you will learn about your options for after custom code analysis.

ABAP Test Cockpit (ATC) Baseline

With the baseline functionality, it's now possible to add/remove custom code project check results to/from the baseline and choose whether you would like to:

Exempt your findings: ATC findings from the baseline will be marked as exempted in the check results list.

Suppress your findings: After a new ATC check run, all ATC findings from the baseline are excluded from the ATC run results list.

Reduce findings' priority: The priority will be reduced to low (priority 3) for all ATC findings of the relevant ATC check result.

Result: After adding findings to the baseline, they won't appear in the subsequent check results of the custom code project.

Note

Please be aware that:

Once you have added findings to the baseline, the custom code migration project cannot be deleted. Remove your findings from the baseline first and then proceed with deleting your project.

The analysis cannot be started again after the findings were added to the baseline.

SAP Readiness Check

You can download the check results to upload them later onto SAP Readiness Check 2.0. To do so, in the Analyze Findings section of your project, choose the arrow next to the spreadsheet icon and select Export for SAP Readiness Check. To learn more about SAP Readiness Check for SAP S/4HANA, see SAP Note 2913617 .

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/programming-languages>

Title: Programming Languages | SAP Help Portal

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Programming Languages

SAP BTP supports many different programming languages; the availability of each depends on the development environment you're using.

The following programming languages are available in the environments:

Supported Environments and Programming Languages

Environment

Programming Language

Cloud Foundry environment

Java

Node.js

Python

ABAP environment

ABAP

Kyma environment

Serverless Functions: Node.js and Python

Microservices: Any cross-platform programming language

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/regions>

Title: Regions | SAP Help Portal

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Regions

You can deploy applications in different regions. Each region represents a geographical location (for example, Europe, US East) where applications, data, or services are hosted.

About SAP BTP Regionsà

Regions are provided either by SAP or by our Infrastructure-as-a-Service (IaaS) partners Amazon Web Services (AWS), Microsoft Azure, Google Cloud, and Alibaba Cloud. The third-party region providers operate the infrastructure layer of the regions, whereas SAP operates the platform layer and Cloud Foundry.

For an overview of all available regions, see SAP Discovery Center:

Selecting a Regionà

A region is chosen at the subaccount level. For each subaccount, you select exactly one region. The selection of a region is dependent on many factors: For example, application performance (response time, latency) can be optimized by selecting a region close to the user. For more information, see [Selecting a Region](#).

Deploying Applications in Regionsà

When deploying applications, consider that a subaccount is associated with a particular region and that this is independent of your own location. You may be located in the United States, for example, but operate your subaccount in a region in Europe. For more information on subaccounts, see [Subaccounts](#).

To deploy an application in more than one region, execute the deployment separately for each host. For more information, see [Deploy an Application](#).

Within a region, there can be multiple instances of the SAP BTP, Cloud Foundry environment. When creating a subaccount, SAP BTP automatically assigns the account to a specific instance of the environment. Several subaccounts of the same global account can be located in different datacenters in one region. This also affects the format of the API endpoint URL that is displayed in the cockpit after enabling Cloud Foundry in your subaccount. There are two possible formats for the API endpoint URL, either displayed with or without an index. Here's an example for eu10:

Example

<https://api.cf.eu10.hana.ondemand.com>

[https://api.cf.eu10-\*\*<XXX>\*\*.hana.ondemand.com](https://api.cf.eu10-<b><XXX></b>.hana.ondemand.com)

In both cases, the subaccount is located in the region eu10. The differences in the URLs are only an indicator of technical details on the side of SAP BTP and do not affect the functionality of your applications. For information on enabling Cloud Foundry, see [Create Orgs](#).

#### High Availability

SAP has a number of processes in place to support resilience in SAP BTP, and provides different offerings so that you can support the high availability of your applications. For more information, see [Resilience](#), [High Availability](#), and [Disaster Recovery](#).

#### EU Access

Some customer contracts include EU Access, which restricts processing of personal data to EEA/Switzerland. If the global account is marked with EU Access, the actual EU Access compliance status of subaccounts will be displayed during creation of subaccounts.

#### Note

If you require a subaccount with EU Access, make sure to select a provider and region where EU Access is available. Regions that allow for EU Access are labeled as such in the map view filter in the SAP Discovery Center.

For some services, EU Access is generally not available, not even if the provider and region support EU Access.

#### Related Information

[Regions and API Endpoints Available for the Cloud Foundry Environment](#)

[Regions and API Endpoints for the ABAP Environment](#)

[Regions for the Kyma Environment](#)

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/resilience-high-availability-and-disaster-recovery>

Title: [Resilience, High Availability, and Disaster Recovery | SAP Help Portal](#)

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#### Resilience, High Availability, and Disaster Recovery

SAP has a number of processes in place to support resilience in SAP BTP, and provides different offerings so that you can support the high availability of your applications.

#### How SAP Provides Resilience

SAP applies resilience principles when developing, updating, and deploying our SAP BTP applications and services.

SAP BTP provides resilience through the following:

##### Processes and Offerings

##### Description

##### Availability Zones

To achieve better fault-tolerance, we deploy our services across multiple availability zones (AZs), which improves the availability of a service if there are issues with the infrastructure of one AZ. Both Kyma and Cloud Foundry utilize multiple AZs provided by the cloud providers to ensure high availability of the platform.

For more information on AZs, see:

Cloud Foundry runtime: [Availability Zones in the Cloud Foundry Environment](#).

Kyma runtime: [Availability Zones in the Kyma Environment](#).

In addition to what SAP BTP offers, you can follow the [Best Practices for Resilient Applications](#) to ensure high availability of your applications.

##### Backups

SAP provides platform-level backups for some foundation services. However, for most of the services, managing backups is the customers' responsibility.

For more information on backups, see [SAP BTP Service Configuration Backups Managed by Customers](#) and [Data Backups Managed by SAP](#).

##### Disaster Recovery

The SAP BTP Disaster Recovery (DR) Plan is part of the overall SAP BTP Business Continuity Plan, which

includes crisis management and process continuity activities that are triggered by a declared disaster. For more information, see Disaster Recovery as Part of the Business Continuity Plan.

#### In-Metro Disaster Recovery

In-Metro DR is a disaster recovery solution that utilizes synchronous data replication. It has been designed to protect our customers from the impact of local disasters that may affect a single availability zone (AZ). By deploying services across multiple AZs within a single region, In-Metro DR ensures that if a disaster occurs in one AZ, the issue is contained within that zone. Meanwhile, the other unaffected AZs continue to function normally, efficiently handling incoming requests. For more information, see Disaster Recovery as Part of the Business Continuity Plan.

#### Best Practices for Resilient Applications

In addition to the services offered by SAP BTP, you can follow our best practices for developing and deploying applications, which allow you to make your application running on SAP BTP stable and highly available.

#### Develop Resilient Applications

When developing your applications, apply the principles and patterns of resilient software design that fit your use case. For more information, see Developing Resilient Apps on SAP BTP. For information specifically about developing applications in the Kyma runtime, see Develop Resilient Applications in the Kyma Runtime. For situations where the load is highly available and where applications need to react by scaling, consider using Application Autoscaler. For more information, see What Is Application Autoscaler?

#### Working with Availability Zones

To benefit from the high availability mechanisms in Cloud Foundry, set up your applications with multiple instances. For more information, see Developing Resilient Applications.

#### Disaster Recovery as Part of the Business Continuity Plan

The cloud platform disaster recovery (DR) plan is part of the overall cloud platform business continuity plan, which includes crisis management and process continuity activities that are triggered by a declared disaster.

#### Standard Disaster Recovery

SAP can restore productive tenants from backups as soon as practicable in case of a disaster resulting in the loss of the primary production data center.

As the magnitude of a disaster is unpredictable, a region might not be restored in a reasonable time. In addition, a new infrastructure might need to be set up at a different location, which might require the purchase and setup of new hardware. Therefore, we can't guarantee any fixed recovery timelines.

#### In-Metro Disaster Recovery

In-Metro DR is a disaster recovery solution that utilizes synchronous data replication. It has been designed to protect our customers from the impact of local disasters that may affect a single availability zone (AZ). By deploying services across multiple AZs within a single region, In-Metro DR ensures that if a disaster occurs in one AZ, the issue is contained within that zone. Meanwhile, the other unaffected AZs continue to function normally, efficiently handling incoming requests.

In-Metro DR is offered to customers of SAP BTP cloud services as contractual obligations and includes:

Recovery Point Objective (RPO): No more than 5 minutes of data loss.

Recovery Time Objective (RTO): Full-service restoration within 2 hours.

In-Metro DR solution leverages an active/active HA set up across 2 or more availability zones with a failover for all customers from primary to secondary site. That is, availability zones within a single region

Fully functioning availability zones are paired with every primary availability zone, application stack is pre-built with same capacity and compute as the primary availability zone.

A written, customer facing DR plan exist and is updated at least once every 12 months.

A DR test is performed at least once every 12 months, and the DR documents are revised as needed based on the results.

The In-Metro DR solution passes annual audits and receives certifications (for example, SOC2).

Scope:

For the scope of SAP BTP services, see SAP BTP Disaster Recovery Overview.

Includes SAP BTP regions on AWS and Azure

For more information, see Announcing the New "In-Metro" Disaster Recovery Solution for SAP BTP.

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/sap-btp-operator-module>

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/sap-btp-specific-configurations>

Title: *SAP BTP-Specific Configurations | SAP Help Portal*

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SAP BTP-Specific Configurations

Learn about technical configurations for Cloud Foundry that are specific to SAP BTP.

The following technical configurations are specific to SAP BTP and differ from the default configuration:

SAP BTP supports the Cloud Foundry command line interface (CF CLI) version 8 or newer. Older versions of the CF CLI are not supported.

By default, a newly pushed (or started) Cloud Foundry application needs to respond to a health check within the first 60 seconds, otherwise the application is considered to have failed. For more information, see [https://docs.cloudfoundry.org/devguide/deploy-apps/healthchecks.html#health\\_check\\_timeout](https://docs.cloudfoundry.org/devguide/deploy-apps/healthchecks.html#health_check_timeout). On SAP BTP, however, you can override this timeout to up to 10 minutes. For instructions, see <https://docs.cloudfoundry.org/devguide/deploy-apps/large-app-deploy.html>.

On SAP BTP, application SSH access is disabled by default. For more information on SSH, see <https://docs.cloudfoundry.org/devguide/deploy-apps/app-ssh-overview.html>.

SAP BTP supports the Cloud Foundry API version 3. The Cloud Foundry API v2 is not available. For more

information, see <https://v3-apidocs.cloudfoundry.org/>.

On SAP BTP, the Cloud Foundry API is protected by a rate limit against misuse. The limit is in the range of a few 10k requests per hour per user on average.

In addition to the general rate limit on the Cloud Foundry API, requests for certain API endpoints related to services face a separate limit on concurrent requests. The Cloud Foundry API responds with HTTP status code 429 if a rate limit is reached and provides a Retry-After Header suggesting when the client can attempt a retry. For more information, see <https://docs.cloudfoundry.org/running/rate-limit-cloud-controller-api.html#Rate%20Limit%20Responses:%20Service%20Brokers>.

In the SAP BTP, Cloud Foundry environment, the total HTTP Request Header and HTTP Response Header size is limited to 64 KB to protect against misuse.

In the SAP BTP, Cloud Foundry environment, for both HTTP Request Headers and HTTP Response Headers the total amount of headers is limited to 101. Components that are handling the requests will add system-relevant headers such as X-Vcap-Request-Id or X-Ssl-\* when mTLS is used. This addition reduces the amount of headers that can be set by the client (for requests) or the application (for responses).

In the SAP BTP, Cloud Foundry environment, the limit of concurrent HTTP connections between client and application is 3000 per application container.

In the SAP BTP, Cloud Foundry environment, the HTTP keep-alive timeout towards the client is set to 60s to protect against misuse. 60s is the maximum time span allowed to wait for a new HTTP request to appear if keep-alive is enabled.

In the SAP BTP, Cloud Foundry environment, an internal HTTP keep-alive is set to 90s. A higher value must be set on application-side to avoid intermittent disruptions. For more information, see step 4 of 3406978 .

In the Cloud Foundry environment, there's a logging rate limit to guard against malicious applications. By default, the limit is 4000 logs per second per application instance, but in exceptional high load scenarios it may be lowered by the platform. If this limit is exceeded, additional logs from the application instance are dropped and a warning message is injected into the application instance's log stream every second. This message also contains the exact current log rate limit.

Applications requiring sent envelopes to be delivered to external Log Management Services should use the Cloud Foundry syslog drain capability. See <https://docs.cloudfoundry.org/devguide/services/log-management.html>.

In the SAP BTP, Cloud Foundry environment, the time between signaling a container to shut down gracefully and forcefully stopping it is set to 60 seconds. The default in Cloud Foundry is 10 seconds, see <https://docs.cloudfoundry.org/devguide/deploy-apps/app-lifecycle.html#shutdown>. This time interval will not be taken into account if there are no explicit kernel signal handlers implemented in the application.

In the SAP BTP, Cloud Foundry environment, applications get a guaranteed CPU share of ¼ core per GB instance memory. As the maximum instance memory per application is 16 GB, this allows for vertical scaling up to 4 CPUs.

If applications running on the same virtual machine don't use their guaranteed CPU, other applications might get more CPU. This isn't guaranteed and might be subject to change in the future. If you encounter performance problems, scale up your application or increase the application start timeout.

The number of running threads per application instance is limited to 10,420. Reaching this limit can cause performance issues.

When pushing or scaling your application, you can define a disk\_quota that can be up to 10 GB. For more information, see <https://docs.cloudfoundry.org/devguide/deploy-apps/manifest-attributes.html#disk-quota>.

When deploying applications on SAP BTP, the maximum application package size is 1.5 GB. If your application is larger than that, the deployment fails. For more information, see <https://docs.cloudfoundry.org/devguide/deploy-apps/large-app-deploy.html>.

In the SAP BTP, Cloud Foundry environment, the hard limit for open file descriptors is 32,768 (32K) per container.

In global accounts that support the consumption-based commercial model you might see a quota limit for certain services. This is a technical limit only, not a business limit. If you need to increase this limit, report an incident to SAP support for component BC-NEO-CIS.

In the SAP BTP, Cloud Foundry environment, the SAP HANA database supports a limited number of simultaneous connections per database. For more information, see [Connecting to the SAP HANA Database in SAP HANA Cloud](#).

In the SAP BTP, Cloud Foundry environment, each application can be mapped to approximately 1000 routes (128 KB). The total length of the routing information must not exceed this limit.

Cloud Foundry Audit Events have a retention period of 14 days. For more information on Audit Events, see <https://docs.cloudfoundry.org/running/managing-cf/audit-events.html>.

To ensure the stable and fair operation of our platform, we enforce rate limiting on container network bandwidth. Specific thresholds are not disclosed to prevent abuse. These limits are carefully set to fully support all expected customer workload.

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/service-binding-parameters>

Title: Service Binding Parameters | SAP Help Portal

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Service Binding Parameters

Some services support additional configuration parameters with the bind request. These parameters are passed in a valid JSON object containing service-specific configuration parameters, provided either in-line or in a file. For a list of supported configuration parameters, see the documentation for the particular service offering.

Note

At the time being user-provided services do not support arbitrary binding parameters.

The SAP Cloud Deployment service supports the following methods for the specification of service-binding parameters:

Method A - In-line parameters into MTA descriptors (deployment or extension descriptors)

Method B - JSON file with external configuration, containing the required service-configuration parameters

A combination of the first two methods, as described in the table below

Note

If service-binding information is supplied both in the MTA's deployment (or extension) descriptor and in a supporting JSON file, the parameters specified directly in the deployment (or extension) descriptor override the parameters specified in the JSON file.

In the MTA deployment descriptor, the requires dependency between a module and a resource represents the binding between the corresponding application and the service created from them (if the service has a type).

For this reason, the config parameter is nested in the requires dependency parameters, and a distinction must be made between the config parameter in the modules section and the config parameter used in the resources section (for example, when used for service-creation parameters).

Service Binding Parameters - method comparison

Method A

Method B

Combination of the two methods

Sample Code

MTA deployment descriptor or extension

âE

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modules:

- name: node-hello-world-backend

type: javascript.nodejs

requires:

- name: node-hdi-container

parameters:

config:

permissions: debugging

Sample Code

A JSON file with the required service-configuration parameters

âE

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modules:

- name: node-hello-world-backend

type: javascript.nodejs

requires:

- name: node-hdi-container

Sample Code

Content of xs-hdi.json

âE

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{

"permissions": "debugging"

}

Sample Code

Additional entry in MANIFEST.MF

âE

ã

Name: xs-hdi.json

MTA-Requires: node-hello-world-backend/node-hdi-container

Content-Type: application/json

Sample Code

âE

ã

modules:

- name: node-hello-world-backend

type: javascript.nodejs

requires:

- name: node-hdi-container

parameters:

config:

permissions: debugging

Sample Code

Content of xs-hdi.json

âE

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{

"permissions": "debugging"

}

Sample Code

Additional entry in MANIFEST.MF

âE

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Name: xs-hdi.json

MTA-Requires: node-hello-world-backend/node-hdi-container

Content-Type: application/json

âf

Note

The examples are only illustrative for service-binding parameters. Each service offering supports its own set of parameters.

Method A shows how to define the service-binding parameters in the MTA deployment descriptor (mtad.yaml).

If you use this method, all parameters under the special config parameter are used for the service-bind request.

This parameter is optional.

Method B shows how to define the service-binding parameters for a service-bind request in a JSON file. Using this method, there are dependencies on entries in other configuration files. For example, when using the JSON method, an additional entry must be included in the MANIFEST.MF file. This new entry has to define the path to

the JSON file that contains the parameters and the name of the resource for which the parameters should be used.

Note that when you use the combination of the two methods, the parameters defined in the descriptor have higher priority than the ones defined in the JSON file.

Note

To avoid ambiguities, the name of the module is added as a prefix to the name of the requires dependency; the name of the manifest attribute uses the following format: <module-name>#<requires-dependency-name>.

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/service-creation-parameters>

Title: Service Instance Parameters | SAP Help Portal

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Service Instance Parameters

Some services support additional configuration parameters in the create-service or update-service request.

These parameters are parsed in a valid JSON object containing the service-specific configuration parameters.

The deploy service supports the following methods for the specification of service instance parameters:

Method A - an entry in the MTA deployment descriptor or the extension

Method B - a JSON file with the required service configuration parameters

Note

If service instance information is supplied both in the deployment (or extension) descriptor and in a supporting JSON file, the parameters specified directly in the deployment (or extension) descriptor override the parameters specified in the JSON file.

Service instance parameters - method comparison

Method A

Method B

Combination of the two methods

Sample Code

MTA deployment descriptor or extension

âE

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resources:

- name: java-uaa

type: com.sap.xs.uaa

parameters:

config:

xsappname: java-hello-world

Sample Code

MTA deployment descriptor or extension

âE

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resources:

- name: java-uaa

type: com.sap.xs.uaa

Sample Code

xs-security.json

âE

ã

{

"xsappname": "java-hello-world"

}

Sample Code

Additional entry in MANIFEST.MF

âE

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Name: xs-security.json

MTA-Resource: java-uaa

Content-Type: application/json

Sample Code

âE

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resources:

- name: java-uaa

type: com.sap.xs.uaa

parameters:

path: <path to directory>

config:

xsappname: java-hello-world

Sample Code

xs-security.json

âE

ã

{

"xsappname": "java-hello-world"

}

Sample Code

Additional entry in MANIFEST.MF

âE

ã

Name: xs-security.json

MTA-Resource: java-uaa

Content-Type: application/json

Using method A, all parameters under the special config parameter are used for the service creation or the service update request. This parameter is optional.

Tip

You can use the optional path parameter to navigate to a JSON file containing service-specific parameters and configurations. The content of the file is used for the creation or update of the service instance.

Note that if you are deploying from a local directory using an mta.yaml, it needs to contain the path to the JSON file, so that the relevant MANIFEST.MF entry is generated.

Note

The path parameter only works when deploying from directory or when it is specified in the development descriptor mta.yaml. It is used only in build time and not during runtime (deployment).

Using method B, there are dependencies on further configuration entries in other configuration files. For example, if you use this JSON method, an additional entry must be included in the MANIFEST.MF file which defines the path to the JSON file containing the parameters as well as the name of the resource for which the parameters should be used.

Note that when you use the combination of the two methods, the parameters defined in the descriptor have higher priority than the ones defined in the JSON file.

Updating Service Instance Parametersâ

Only valid for service instances managed by service brokers

Updating service instance parameters is, by default, fail-safe for those operations which result in a call to the service brokers. You can control this behavior by using the fail-on-service-update parameter.

Possible values for the parameter are:

null (default value): If the update triggers an asynchronous operation (Cloud Controller sends the parameters to the service broker), the deployment will not fail even if the operation fails.

true: If the update of the service parameters fails (even asynchronously), the deployment will also fail.

false: The update becomes even more permissive - even if the initial call to update the service parameters fails,

the deployment will still continue.

Example

âE

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parameters:

fail-on-service-update:

parameters: true

Only valid for user-provided service instances

Related Informationâ^

Standalone Application Router - HTML5 Application Repository

Managing Service Instances with the cf CLI

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/set-target-for-subsequent-commands-with-btp-target>

Title: Set a Target for Subsequent Commands with btp target | SAP Help Portal

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Set a Target for Subsequent Commands with btp target

Set the target for command calls to a subaccount, a directory, or a global account with the btp target command.

Contextâ^

During login, the btp CLI asks you for the global account you want to log in to and this global account is then the target. This means that commands are executed in this global account unless you specify otherwise via a parameter. There are two typical use cases for the target command:

If you know that you need to work in a particular subaccount or directory, you can change the target, so that you won't have to specify that subaccount's or directory's ID as a parameter with every command call.

If you want to switch to another global account, you can do so by setting the target to this global account without logging in again.

By setting the target to a subaccount or directory, you also target its parent entities, that is, the target hierarchy.

If no parameter is specified in a command call, the btp CLI will execute the command in the innermost targeted entity where a command is applicable. If a subaccount is targeted that is inside a directory and the command is not available on subaccount level, it will be executed in the parent directory (if it's a directory command) or in the parent global account (if it's a global account command). This can be useful, for example, if you have targeted a subaccount, but want to update its parent directory. You can then execute btp update accounts/directory without specifying the directory ID.

To see the current target, use btp --info or simply btp.

To explicitly execute a command in a parent entity instead of in the target, you can specify this with '-dir' or '-ga' parameters without a value. The value is then taken from the target hierarchy. This can be useful, for example, if you have targeted a subaccount, but want to list the users of the parent directory. You can then execute btp list security/user -dir. Note, however, that this command is only available on a directory level if the directory is enabled to manage authorizations.

Usage: btp [OPTIONS] target [PARAMS]

Procedureâ^

Use btp target to set the target for subsequent commands.

Run the command without parameters to see the children, siblings and parent of you current target. You can then select your target or navigate up and down.

Enter --hierarchy as parameter to see all your global accounts with their entire account hierarchies. You can then select your target from this list.

Tip

If you like the --hierarchy parameter and would like to make it the default for the target command, you can define it as a configuration setting with btp set config --target.hierarchy true. If you then run btp target, the entire hierarchy of all your global accounts is shown.

Specify the subaccount, directory, or global account you want to target as described in the next step.

Enter one of the following parameters:

## Parameter

### Description

--global-account, -ga <SUBDOMAIN>

The subdomain of the global account to be targeted.

--directory, -dir<ID>

The ID of the directory to be targeted. You can find the directory ID by using `btm get accounts/global-account --show-hierarchy`.

--subaccount, -sa <ID>

The ID of the subaccount to be targeted. You can find the subaccount ID by using `btm list accounts/subaccount`.

### Results

The CLI client displays the current target as a hierarchy. To execute a command in the targeted entity, you can omit the corresponding parameter.

### Example

A global account can group together different directories and subaccounts that the global account administrator makes available to users. CLI commands can be executed on all levels of this account hierarchy, that is in the global account, a directory, or a subaccount, usually specified by the above-mentioned parameters. The table below shows some example commands, and explains in which account entity they can be executed and how the target mechanism can be used.

Let's look at a global account that contains a directory with a targeted subaccount. With `btm`, the current target hierarchy is displayed like this:

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Current target:

My Global Account (global account, subdomain: cee12xx112345-ga)

% % My Directory (directory, ID: 371eXXXX-55XX-40XX-b3XX-e9947ed9XXXX)

% % My Subaccount (subaccount, ID: d8aeXXXX-74XX-49XX-89XX-f058029eXXXX)

âf

Now, when you run commands without specifying an account entity as parameter, they are executed in the targeted subaccount or in one of its parents.

### Example Command

Command is Available for

Command is Executed in

`btm get accounts/available-environment`

--subaccount

The targeted subaccount

`btm update accounts/directory`

[directory] ID

Note that the directory ID is a positional parameter in this command. You can omit it when a subaccount inside the directory is targeted: `btm update accounts/directory [directory ID]`.

The parent directory of the targeted subaccount

`btm list accounts/available-region`

--global-account

The parent global account of the targeted subaccount

`btm list security/user`

--global-account

--directory

--subaccount

The targeted subaccount

### Tip

To execute this command in a parent entity, use the `-dir` or `-ga` parameter without a value.

`btm list security/user --subaccount "1111XXXX-22XX-33XX-44XX-55555555XXXX"`

--global-account

--directory

--subaccount

The specified subaccount, which overrides the target

Related Informationà

btp CLI Command Reference

Global Accounts

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/tools-programming-models-programming-languages-and-apis>

Title: Tools, Programming Models, Programming Languages, and APIs | SAP Help Portal

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Tools, Programming Models, Programming Languages, and APIs

SAP BTP provides various programming languages and tools for your development project.

Tools

SAP BTP includes many tools to help you develop and manage applications, and connect them to your on-premise systems.

Programming Languages

SAP BTP supports many different programming languages; the availability of each depends on the development environment you're using.

Programming Models

Continuous Integration and Delivery (CI/CD)

Configure and run predefined continuous integration and delivery (CI/CD) pipelines that automatically build, test, and deploy your code changes to speed up your development and delivery cycles.

APIs

Discover and consume APIs to manage, build, and extend the core capabilities of SAP BTP.

Cloud Management Tools

Cloud management tools represent the group of technologies designed for managing SAP BTP.

Prerequisites and Restrictions

Find a list of the product prerequisites and restrictions for SAP BTP.

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URL: <https://help.sap.com/docs/btp/sap-business-technology-platform/what-is-consumption-based-commercial-model>

Title: What Is the Consumption-Based Commercial Model? | SAP Help Portal

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What Is the Consumption-Based Commercial Model?

Note

The use of the consumption-based commercial model is subject to its availability in your country or region.

With the consumption-based model, your organization purchases an entitlement to all current and future SAP BTP services that are eligible for this model. Throughout the duration of your contract, you have complete flexibility to turn services on and off and to switch between services as your business requires.

The consumption-based commercial model is available in the following flavors: the SAP BTPEA (SAP BTP Enterprise Agreement), the CPEA (Cloud Platform Enterprise Agreement), and Pay-As-You-Go for SAP BTP. Each option is suited to different business situations and levels of financial commitment, as described in the table below. For additional information and clarifications, please contact your account executive.

SAP BTPEA

Your organization makes a prepaid investment in cloud credits for the contract duration with an annual commitment to consume SAP BTP services.

This model is suitable for customers that have well-established and planned use cases, and who want the flexibility of turning services on and off, and switching between services, without the commitment of being tied to a single service throughout the duration of the contract.

You receive a monthly balance statement that provides information about the usage consumption of each service and the corresponding costs. The total monthly cost is deducted from your cloud credits balance. This model has a minimum investment entry, and volume-based discounts are available. You are billed annually in advance. Any overages are billed in arrears at list price. You can top up your cloud credits at any time to prevent overages. You get access to new SAP BTP services that are added to the consumption-based service catalog.

#### CPEA

##### Note

Certain restrictions apply. See the Frequently Asked Questions section in <https://www.sap.com/products/technology-platform/pricing.html>.

Your organization makes a prepaid investment in cloud credits for the contract duration with an annual commitment to consume SAP BTP services.

This model is suitable for customers that have well-established and planned use cases, and who want the flexibility of turning services on and off, and switching between services, without the commitment of being tied to a single service throughout the duration of the contract.

You receive a monthly balance statement that provides information about the usage consumption of each service and the corresponding costs. The total monthly cost is deducted from your cloud credits balance.

This model has a minimum investment entry, and volume-based discounts are available.

You are billed annually in advance. Any overages are billed in arrears at list price.

You can top up your cloud credits at any time to prevent overages.

#### Pay-As-You-Go for SAP BTP

You have the same access to all the services that are available in SAP BTPEA, but with a highly flexible zero-commitment model – you pay nothing upfront and there is no minimum usage requirement or annual commitment.

You pay only for the SAP BTP services that you want, when you use them.

You are billed monthly in arrears.

Service charges are non-discountable.

This low-risk model is suitable for customers with use cases that are not well defined, and are interested in running a proof-of-concept in a productive environment. This model provides the flexibility of turning services on and off, and switching between services, as needed throughout the duration of the contract.

A seamless transition to the CPEA model is available, on the condition that you have no other CPEA-based global accounts.

If you want information about the Pay-As-You-Go for SAP BTP for cloud test, demo, and development commercial model, go to the SAP Partner Portal.

##### Note

In global accounts that use the consumption-based commercial model, a default runtime memory quota of 200 GB for the standard plan is automatically assigned to every subaccount. This quota represents a limit on how much runtime memory the applications in the subaccount can use at any given time. When you use the standard plan for SAP BTP, Cloud Foundry runtime in the consumption-based commercial model, you're billed based on the amount of runtime memory that has actually been consumed. For more information, see [Service Plans and Metering for Cloud Foundry Runtime and Consumption Monitoring](#).

A global account administrator can change the default quota as described in (optional) [Adjusting Runtime Memory Quota for Cloud Foundry Runtime \(Consumption-Based\)](#).

##### Tip

You can monitor costs and service usage throughout the contract period. See [Monitoring Usage and Consumption Costs in Your Global Account](#).

For information about eligible services and pricing, see <https://www.sap.com/products/technology-platform/pricing.html>, or access the SAP BTP service catalog via the SAP Discovery Center. The SAP BTP service catalog allows you to identify service availability per data center and to determine licensing model compatibility per available service plan.

As part of the free tier model for SAP BTP, some services offer free service plans that allow you to try out these services without any additional charge within your SAP BTPEA, CPEA, or Pay-As-You-Go for SAP BTP global account. For more information, see [Using Free Service Plans](#).

To find frequently asked questions about the consumption-based commercial model, see the [Frequently Asked](#)

Questions section in <https://www.sap.com/products/technology-platform/pricing.html> .

#### Note

Depending on your business needs, you can combine the consumption-based commercial model with the subscription-based commercial model in the same global account, or you can use both commercial models in the same global account. Contact your SAP account executive or sales representative for more information.

Note however that you cannot mix more than one flavor of the consumption-based commercial model in the same global account.

#### Related Information^

What Is the Subscription-Based Commercial Model?

Monitoring Usage and Consumption Costs in Your Global Account

View Subaccount Usage Analytics

Managing Global Accounts Using the Cockpit

Managing Directories Using the Cockpit

Managing Subaccounts Using the Cockpit

Managing Entitlements and Quotas Using the Cockpit

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